84

Care of the Older Adult With Chronic Kidney Disease

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CHAPTER OUTLINE

KEY POINTS

- Common equations used to estimate the glomerular filtration rate (GFR) have wider margins of error when applied to older adults. These may misclassify individuals as having chronic kidney disease (CKD), resulting in debate about whether it is appropriate to label individuals, in their eighth and ninth decade, as having a chronic disease based on current criteria.
- Older individuals with CKD appear to have an unexpectedly high burden of geriatric syndromes (GSs) such as frailty, cognitive impairment, functional limitation, depression, incontinence, sensory deficits, malnutrition, and falls.
- Patients with CKD often use polypharmacy. Polypharmacy in older adults is further complicated by the altered pharmacokinetics and pharmacodynamics in the setting of advancing age and presence of CKD.
- A complete evaluation (known as a comprehensive geriatric assessment; CGA) is recommended for all older individuals. This should include a detailed medical history (including treatment targets); assessment of the home environment and social support structure, functional history, and assessment of personal values and lifestyle preferences.
- Although it is often assumed that treatment with dialysis will extend life and alleviate the signs and symptoms of advanced kidney disease, there is growing evidence that these benefits may not always accrue in older adults.
- Several studies have suggested that dialysis initiation may exacerbate or accelerate functional and cognitive decline, resulting in a 1.6-fold higher risk of serious fall injuries and a higher risk of requiring professional nursing care.
- It is important to distinguish among younger and older individuals, robust or frail, and those with renal-limited CKD and those with complex clustering of multiple chronic diseases while engaging in shared decision-making processes.
- Most studies have shown that dialysis is associated only with a modest survival advantage over comprehensive conservative renal care.

INTRODUCTION

Although chronic kidney disease (CKD) is a significant public health issue for patients of all ages, there is a limited body of literature addressing the management of CKD in older adults. Most health care practitioners would agree, however, that caring for older adults with CKD presents unique challenges because of the substantial differences between older and younger patients with this condition. In this chapter, we will provide an overview of a broad number of issues related to providing care for the older adult with CKD. We have included discussions about the emerging debate rejecting current definitions used to define CKD, detailed information about common geriatric syndromes (e.g., falls, functional decline, cognitive impairment) and their relationship with renal disease and treatment, and introduced the role of palliative dialysis and comprehensive conservative renal care (CCRC).

CHRONIC KIDNEY DISEASE AND AGE-RELATED CHANGES IN ESTIMATED GLOMERULAR FILTRATION RATE

Kidney Disease Improving Global Outcomes (KDIGO) clinical guidelines define CKD based on measured serum creatinine level and estimated glomerular filtration rate (eGFR). Traditionally, the presence of an eGFR of less than 60 mL/min/1.73 m^2 for longer than 3 months is considered consistent with CKD. Using this definition, CKD prevalence in those aged 65 years and older is approximately 10-fold higher than that seen in those younger than 50 years (40% vs. 4%, respectively).[1] Estimates of prevalence, however, vary widely from study to study and appear to be highly dependent on the method used to evaluate renal function and on the population under study. One uniform observation, however, is that older age is associated with lower renal function than that seen in younger patients. Using creatinine-based estimates, more than 75% of the older population identified as having kidney disease fall into the CKD stage 3a category (eGFR = 45–59 mL/min/1.73 m^2).[1,2]

The high prevalence of CKD in older adults is attributed to multiple factors, including improved longevity despite chronic diseases such as diabetes, hypertension, and cardiovascular disease, newer population screening strategies and kidney damage resulting from newer drugs or medical interventions used for revascularization, and cancer management or organ failure.[3] The rate of renal decline varies among individuals. In general, however, older individuals have a lower rate of renal function decline than younger people, particularly at lower levels of eGFR. In an observational study of 4562 patients older than 65 years of age, Arora et al. have found that 62% of patients with CKD stage 3b and 69% of patients with CKD stage 4 progressed at an annual rate of <1 mL/min/1.73 m^2.[4] Similar findings have been reported elsewhere[5,6] and likely reflect the fact that relatively few older patients with an eGFR < 45 mL/min have albuminuria.[7,8]

EVALUATION OF RENAL FUNCTION IN OLDER ADULTS

In most cases, kidney function in routine clinical practice is estimated based on serum creatinine or cystatin C concentration, rather than determined using exogenous filtration markers such as inulin, iothalamate, or other radiocontrast clearance measures. This is discussed in detail in Chapter 23. Formal evaluation of renal function, using inulin or iothalamate, is impractical and costly in the clinical setting. Similarly, 24-hour urine collections can be cumbersome for older individuals to collect and are more likely to be unreliable due to inappropriate collection, incontinence, or accidental spills. Current KDIGO guidelines[9] recommend reporting the eGFR using the 2009 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine-based equation with the use of additional tests such as cystatin C or a clearance measurement for confirmatory testing only in specific circumstances. European guidelines, specific to the older population,[10] allow more flexibility and do not recommend one equation over the other. Neither guideline, at present, distinguishes among different age groups, or among robust, vulnerable, and frail individuals. In older adults, however, the margins of error associated with eGFR estimates may be wider than in younger populations, and commonly used formulae may overestimate or underestimate the true GFR in older adults. Although beyond the scope of this chapter to give a detailed discussion of the attributes of the various methods for GFR estimation, several clinical factors can affect serum creatinine and cystatin C levels, independent of the GFR. These non-GFR determinants are more common among older rather than younger individuals. They include, among other factors, a low dietary intake of protein and reduced muscle mass (both factors specific to creatinine levels), subclinical inflammation, and a change in body habitus with increased adiposity. Even in the absence of overt disease, older individuals are more likely to be frail or have chronic inflammation arising from normal age-related conditions (sometimes referred to as “inflammaging”). This causes a systematic bias, resulting in wider margins of error when commonly used formulae are applied to older adults. Use of the CKD-EPI equation[11] has superseded the Modification of Diet in Renal Disease (MDRD) Study equation[12] in many parts of the world because it is less likely to misclassify individuals as having CKD[13]—although it is unclear if the improved accuracy is clinically meaningful in older adults. Other formulae, including those using cystatin C[14–16] and those specific to the older population, have been recently proposed but further validation, in external data sets, is awaited (Table 84.1).

Albuminuria is known to be an important clinical finding that is associated with increased mortality, vascular events, and progression to end-stage renal disease (ESRD) across all ages and levels of eGFR.[17–19] In the Berlin Initiative Study (BIS), the albuminuria rate among 2069 adults 70 years of age and older was 26%. Unlike younger individuals, only a small percentage had macroalbuminuria (3.6% had an albumin-to-creatinine ratio [ACR] ≥ 300 mg/g).[20] Similar results were found in the National Health and Nutrition Examination Survey (NHANES) III population, where 21% presented with albuminuria at 70 years of age and older and 32.7% at 80 years and older.[21] The prevalence of albuminuria in older individuals seems substantially lower than that of abnormal eGFR; more than 50% of the 75+ years group are diagnosed with CKD based on an eGFR < 60 mL/min/1.73 m^2 in the absence of albuminuria (Fig. 84.1).[22] This further adds doubt to current definitions of CKD in older patients that

Table 84.1 Estimated Glomerular Filtration Rate (eGFR) Equations

Research Group	Endogenous Filtration Marker (units)	Equation	Development Population	Limitations
Cockcroft-Gault[301]	Creatinine (mg/dL)	$CrCl = [(140 - age) \times weight) / (72 \times S_{Cr})] \times 0.85$ (if female)	N = 505 (236 for training sample) whites, 4% women hospitalized pts Age range: 18–92 yr (only 17 pts > 80 yr)	Values not adjusted for BSA Developed to predict CrCl instead of GFR Underestimates GFR in older adults, especially at higher GFRs
MDRD[302]	Creatinine (μmol/L)	$eGFR = 175 \times (S_{Cr})^{-1.154} \times (age)^{-0.203} \times (0.742 \text{ if female}) \times (1.212 \text{ if black})$	N = 1628 whites and blacks; 40% women Nondiabetic pts with CKD Mean age: 51 ± 13 yr (age range, 18–70 yr)	Tends to underestimate renal function in those with a normal eGFR Not validated in pts >70 yr, at extremes of body weight, and in some ethnic groups (e.g., Asian ethnic groups)
CKD-EPI[11,303]	CKD-EPIcr—creatinine (mg/dL)	$eGFR = 141 \times \min(S_{Cr/\kappa,1})^{\alpha} \times \max(S_{Cr/\kappa,1})^{-1.209} \times 0.993^{age}$ [×1.018 if female] [×1.159 if black] where κ is 0.7 for females and 0.9 for males	N = 8254 nonblacks and blacks, 44% women Mean age: 47 ± 15 yr (only 216 pts > 70 yr and 41 > 80 yr)	Developed in a population with few pts > 70 yr and almost none > 80 yr Less accurate in men with BMI ≥ 35 kg/m^2
	CKD-EPIcys—cystatin C (mg/L)	$eGFR = 133 \times \min(S_{cys/0.8,1})^{-0.499} \times \max(S_{cys/0.8,1})^{-1.328} \times 0.996^{age}$ [×0.932 if female] where min indicates the minimum of $S_{Cr/\kappa}$ or 1 and max indicates the maximum of $S_{Cr/\kappa}$ or 1	N = 5352 nonblacks and blacks; 42% women Mean age: 47 ± 15 yr (719 pts > 65 yr)	
LMR[304]	Creatinine (μmol/L)	$eGFR = e^{X - 0.0158 \times age + 0.438 \times \ln(age)}$ Female: pCr < 150 μmol/L: $X = 2.50 + 0.0121 \times (150 - pCr)$ Female: pCr ≥ 150 μmol/L: $X = 2.50 - 0.926 \times \ln(pCr/150)$ Male: pCr < 180 μmol/L: $X = 2.56 + 0.00968 \times (180 - pCr)$ Male: pCr ≥ 180 μmol/L: $X = 2.56 - 0.926 \times \ln(pCr/180)$	N = 3495 Swedish whites; 47% women Mean age: 63 yr (age range, 21–86 yr; 756 pts 70–79 yr and 342 pts ≥ 80 yr)	Difficult to apply Overestimation in pts with BMI < 20 kg/m^2, especially males Not validated in other racial and ethnic groups (e.g., black, Asian) Lack of external validation
FAS[305,306]	Creatinine (mg/dL)	$eGFR = \dfrac{107.3.}{(SCr/Q_{crea})} \times 0.988^{(age-40)}$ Equation for age > 40 yr, where Q_{crea} = 0.7 mg/dL for females and 0.9 mg/dL for males	N = 6870 nonblacks; 47% women Mean age: 74 yr (1764 pts ≥ 70 yr).	May perform better in healthy and general population than in the CKD population Needs further external validation (e.g., blacks, Asians) Clinical utility needs to be confirmed in further studies
	Cystatin C (mg/L)	$eGFR = \dfrac{107.3.}{(S_{cysC}/Q_{cysC})} \times 0.988^{(age-40)}$ where Q_{cysC} = 0.82 mg/L when age 40–70 yr and 0.95 mg/L when age ≥ 70 yr	N = 6132 nonblacks; 7% women Mean age: 58 ± 18 yr (1469 pts ≥ 70 yr)	
BIS[307]	BIS1—creatinine (mg/dL) BIS2—cystatin C (mg/L)	BIS1: $3736 \times creatinine^{-0.87} \times age^{-0.95} \times 0.82$ (if female) BIS2: $767 \times cystatin\ C^{-0.61} \times creatinine^{-0.40} \times age^{-0.57} \times 0.87$ (if female)	N = 570 (for equation development; whole cohort, N = 2073) White German participants; 43% women Community-dwelling older adults ≥ 70 yr Mean age: 78 yr (208 pts ≥ 80 yr)	Needs further external validation (e.g., blacks, Asians)

BIS, Berlin Initiative Study equation; *BMI,* body mass index; *BSA,* body surface area; *CKD-EPI,* Chronic Kidney Disease-Epidemiology Collaboration equation; *FAS,* Full Age Spectrum equation; *LMR,* revised Lund-Malmö equation; *MDRD,* Modification of Diet in Renal Disease Study equation; *pts,* patients.

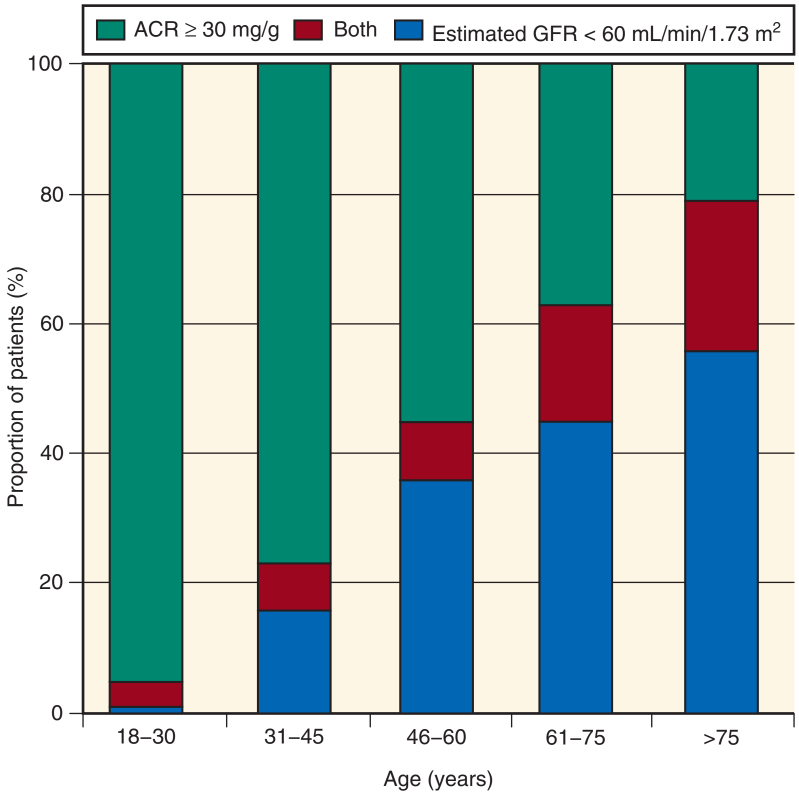

Fig. 84.1 Proportions of patients with chronic kidney disease identified by the albumin-to-creatinine ratio (ACR), estimated glomerular filtration rate (GFR), or both in the US population. (Adapted from James, MT, Hemmelgarn BR, Tonelli M. Early recognition and prevention of chronic kidney disease. *Lancet.* 2010;375(9722):1296–1309.)

only include a low eGFR without consideration of albuminuria or proteinuria.

In addition to the relationship between albuminuria and vascular health, there is a strong association between the presence of CKD and vascular events.[17,23–25] In the Cardiovascular Health Study, a community-based cohort of US adults aged 65 years and older, individuals with CKD had more than twice the prevalence of coronary artery disease compared with those without CKD.[26,27] In three large cohorts, the Kidney Early Evaluation Program (KEEP; N = 27,017), NHANES, 1999 to 2006 (N = 5,538), and the Medicare 5% sample (N = 1,236,946), the prevalence of diabetes in those with CKD ranged from 21.4% to 46.2% and that of hypertension was 90% or higher.[25] High rates of concomitant high cholesterol, coronary artery disease, congestive heart failure, cardiovascular disease, and cancer were also seen in the KEEP and NHANES cohorts (Fig. 84.2). This relationship remained consistent across age groups.

The lower rate of renal decline observed in older individuals, especially those older than 75 years, means that older adults are more likely to die than to develop ESKD, even when eGFR is severely reduced. In a large Veterans Administration (VA) cohort, the risk of death exceeded that of dialysis in adults older than 85 years, regardless of eGFR (Fig. 84.3).[5] In a community-dwelling population of 1268 older individuals (mean age, 75 years) with an eGFR < 60 mL/min/1.73 m^2 followed for 9.7 years, death was 13 times more likely than progression to ESKD, and 25 times more likely for those between 76 and 85 years of age. Even among those who had more advanced CKD (eGFR < 45 mL/min/1.73 m^2), the risk of death remained more than sixfold higher than the risk of ESKD.[28] Clinical prediction models to assess risk of death among older patients with CKD are rare and not widely used.[29–31] Instead, prediction models that predict the likelihood that an individual will progress to ESKD are increasingly being used in clinical practice. Using two large Canadian cohorts of patients with CKD stages 3 to 5 defined using eGFR, Tangri et al. developed, and subsequently validated in diverse global cohorts (N = 721,357), two risk equations, termed the eight- and four-variable kidney function risk equation (KFRE).[32,33] The more popular four-variable KFRE uses age, gender, eGFR, and urinary ACR and performs sufficiently well for use in older patients with an eGFR < 45 mL/min/1.73 m^2.[10] One criticism of the four-variable KFRE is that it does not address the risk, particularly in older populations, that an acute illness may be associated with an acute precipitous drop in renal function. In fact, between 25% and 50% of those starting dialysis do so after an episode of acute kidney injury (AKI)[34,35] and it has been suggested that future modeling equations include the patient's competing risk of death prior to dialysis requirements, rate of renal function decline, and information about the AKI risk.[36]

NORMAL AGING VERSUS DISEASE: IS THERE A NEED FOR CHANGE IN HOW WE DEFINE CHRONIC KIDNEY DISEASE?

Inaccuracies around eGFR estimation, and the observation that older patients have less albuminuria and slower deterioration rates, have been fueling debate about whether it is appropriate

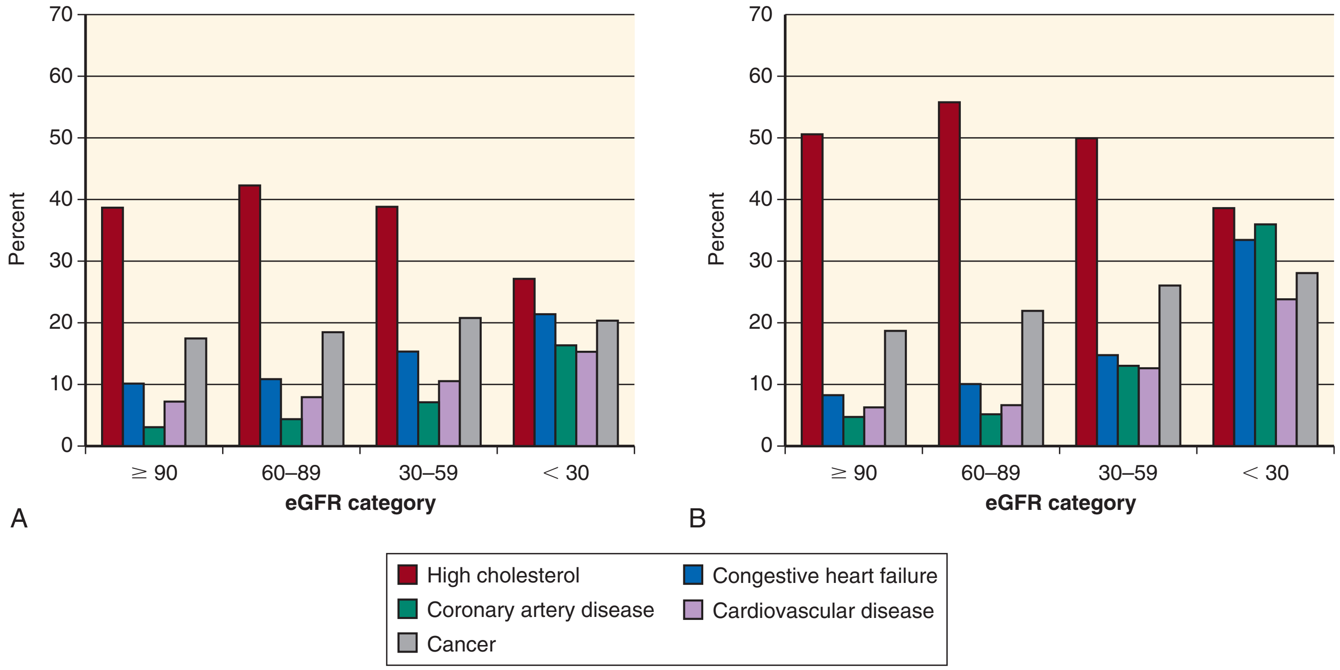

Fig. 84.2 Comorbid conditions by level of estimated glomerular filtration rate (eGFR) in the Kidney Early Evaluation Program (A) and National Health and Nutrition Examination Survey (B). (From Stevens LA, Li S, Wang C, et al: Prevalence of CKD and comorbid illness in elderly patients in the United States: results from the Kidney Early Evaluation Program (KEEP). *Am J Kidney Dis.* 2010;55(Suppl 2):S23–S33.)

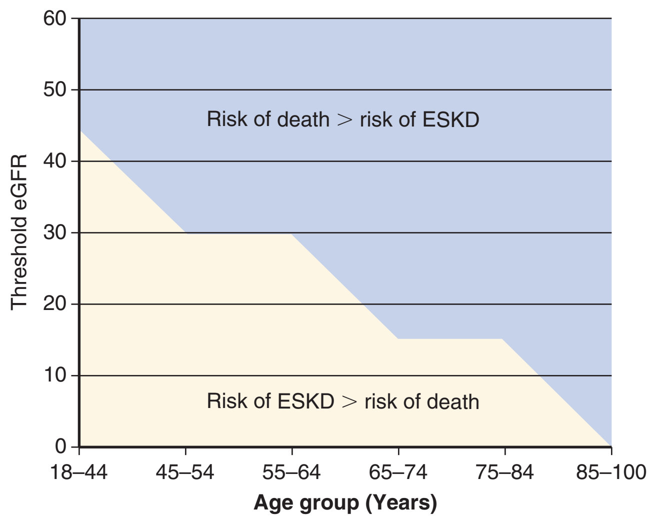

Fig. 84.3 Baseline estimated glomerular filtration rate (eGFR) threshold below which the risk for end-stage kidney disease (ESKD) exceeded the risk for death for each age group. (From O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol.* 2007;18:2758–2765.)

to label individuals, in their 8th and 9th decades, as having a chronic disease based on current criteria. More recently, several authors have suggested that the prevalence of CKD is greatly overestimated in older individuals.[37,38] They note that a decline in renal function is a normal component of aging and suggest that the nomenclature used to define CKD be modified, and based on age.[39–41] Contemporary definitions of CKD do not differentiate between older and younger adults and likely result in a number of older individuals being mislabeled and potentially overtreated for CKD.[42–45]

Several studies have suggested that the GFR declines steadily with healthy aging, often at a slow rate and in the absence of albuminuria. Although these changes are often attributed to fibrotic changes in the kidney, there is insufficient evidence to suggest that pathologic findings parallel changes in renal function. In an evaluation of 1203 healthy potential kidney donors, Rule and colleagues[46] have correlated the degree of renal fibrosis on histology with the GFR measured using iothalamate clearance across a number of age groups. This reduction in renal function appeared to begin in the 3rd decade, but was most discernible by the 6th or 7th decade.[47] Fibrosis was more common with advancing age, and the measured GFR fell with age (estimated at 4.6 mL/min/1.73 m^2 per decade in men and 7.1 mL/min/1.73 m^2 per decade in women), but there was substantial heterogeneity in GFR values among patients with similar degrees of fibrosis and of the same age, and the resulting correlation between the GFR and degree of fibrosis was weak.[48] (See Chapter 22 for further discussion on the physiology and pathophysiology of the aging kidney.)

COMORBIDITIES AND GERIATRIC SYNDROMES AMONG OLDER INDIVIDUALS WITH CHRONIC KIDNEY DISEASE

"Geriatric syndromes" (GSs) is the term used to describe a group of common health conditions in older people that do not fit into discrete disease categories. These conditions include frailty, functional limitation, falls, depression, polypharmacy, malnutrition, and cognitive impairment. Collectively, the GSs arise from a complex interplay between age-related physiologic changes, chronic disease, and functional stressors in older adults. Each interacts with the other, and often the syndromes co-occur.

Emerging data in older individuals with CKD have suggested an unexpectedly high burden of GSs. It is important for nephrology teams to recognize these geriatric issues for several reasons. The appropriateness and effectiveness of medical interventions, such as dialysis care, are dependent on the overall patient well-being, GSs, and general health. Moreover, from the patient's perspective, the presence of one or more GSs almost invariably affects their lifestyle, as well as that of their family and social circle. Timely recognition of these syndromes can help with treatment decision making, and trigger discussions around goals of care and advance care planning.

The formal evaluation of a patient, using a geriatric lens to recognize GSs and incorporate findings into clinical care, is referred to as a comprehensive geriatric assessment (CGA). Ideally, the CGA assessment is conducted over several visits, in the home and in the clinic, by a variety of health care professionals in addition to the physician, including nursing, physiotherapy, occupational therapy, and social work. Evaluation includes a detailed medical history, including treatment targets, assessment of the home environment and social support structure, functional history, and an assessment of personal values and lifestyle preferences. This information is then coupled with the medical issues and assessment of strength, balance, sensory function, cognition, depression, nutrition, and communication skills and used to identify potential areas of concern. Multidisciplinary teams use the results of the assessment to develop a coordinated plan that maximizes health while minimizing the impact of the geriatric impairments, or the treatments being initiated, on the individual and his or her caregivers. This multidimensional and multidisciplinary approach to GSs has been shown to be effective in the general population[49–51] and across multiple renal clinical settings.[52–55] Awareness of GSs can improve outcomes and well-being,[52,55] and routine CGA has been recommended by the American Society of Nephrology, Geriatric Advisory Group, for all older CKD patients.[56]

FRAILTY

In the clinical realm, frailty is a complex syndrome (or state) associated with increased vulnerability. It often results from a combination of problems occurring across different domains of daily functioning, including physical, sensory, psychological, and social domains. Frailty has multiple causes and contributors and is characterized by diminished strength and endurance and by reduced physiologic function that leave the individual vulnerable to developing increased dependency and/or death.[57,58] Operational definitions for determining frailty vary, and a multitude of validated measurement scales have been used.[59] This contributes to much of the variability in estimated prevalence. The sole use of the physical domain as a surrogate measure for the global assessment of frailty, such as the Fried et al. classification,[60] in which definitions are based on five physical components—unintentional weight loss, decreased strength, decreased exercise tolerance, reduced gait speed, and fatigue, may be an inadequate metric for frailty, particularly in CKD. Social determinants of frailty, such as those associated with environmental and mental well-being, are also of importance and may be recognized through more holistic assessments.[61,62] The Clinical Frailty Scale is an example of a simple and clinically applicable validated tool that can be used to assess global frailty status. The scale uses several grades of frailty (Fig. 84.4) based largely on the clinical judgment of the physician, making it an easily applied screening instrument to identify those who would most benefit from a formal CGA.

Frailty is of clinical importance because it is associated with an increased risk of both morbidity (e.g., falls, hospitalization, dependence, need for long-term care) and mortality.[63–65] Risks of mortality are twice as high in those with frailty, compared with nonfrail individuals, across the complete spectrum of CKD. For example, in CKD patients enrolled in the MDRD study, frailty determined from self-reported questionnaires was associated with a 1.7-fold increase in mortality over time (hazard ratio (HR) 1.7; 95% confidence interval [CI], 1.3–2.3)[66] and, in frail patients starting dialysis in a special US Renal Data System (USRDS) study, mortality was 1.8-fold higher over a 3-year period (HR, 1.8; 95% CI, 1.4–2.2).[65]

The prevalence of frailty appears to increase as renal function declines[66,67] and is estimated to be two- to fourfold higher in the renal population, compared with those with normal renal function.[67–69] Recent prevalence estimates have suggested that between 40% and 60% of those on dialysis have frailty characteristics.[63,70,71] Although more prevalent in those with increased age, there is also an unexpectedly high burden of frailty in younger individuals on dialysis.[72,73] This high prevalence may be partly explained by concomitant CKD-related conditions such as protein-energy wasting (PEW), anemia, inflammation, acidosis, and hormonal disturbances.[74] There is also an increasing awareness of how renal care and treatments may inadvertently promote the sick role and create barriers to routine participation in exercise programs and other positive health behaviors.[63]

Frailty is a dynamic process in which transitions across states of frailty are common but, on average, transition to a worse state is more common than to a better state.[75–77] Although characterized by a decline in physical function, including physical activity and exercise capacity, it is often also associated with reduced social participation and self-care abilities, depression, and cognitive impairment and should trigger open discussions reviewing the current goals of care, prognosis, and more frequent evaluation of symptoms. To date, there are no simple one-step interventions that reverse frailty, but multidisciplinary care, nutritional supplementation, exercise, and rehabilitation may attenuate the morbidity associated with frailty, particularly when applied early.[24,74]

FUNCTIONAL IMPAIRMENT

Functional status describes an individual's ability to perform tasks associated with personal care and maintaining a household. In contrast to leisure activities (e.g., gardening, sports), functional tasks are often regarded as fundamental for daily life, and loss of independence is associated with reduced quality of life (QOL) for patients and their caregivers. Although patients with frailty characteristics have a high burden of functional dependence, both frailty and functional loss may occur independently of each other.

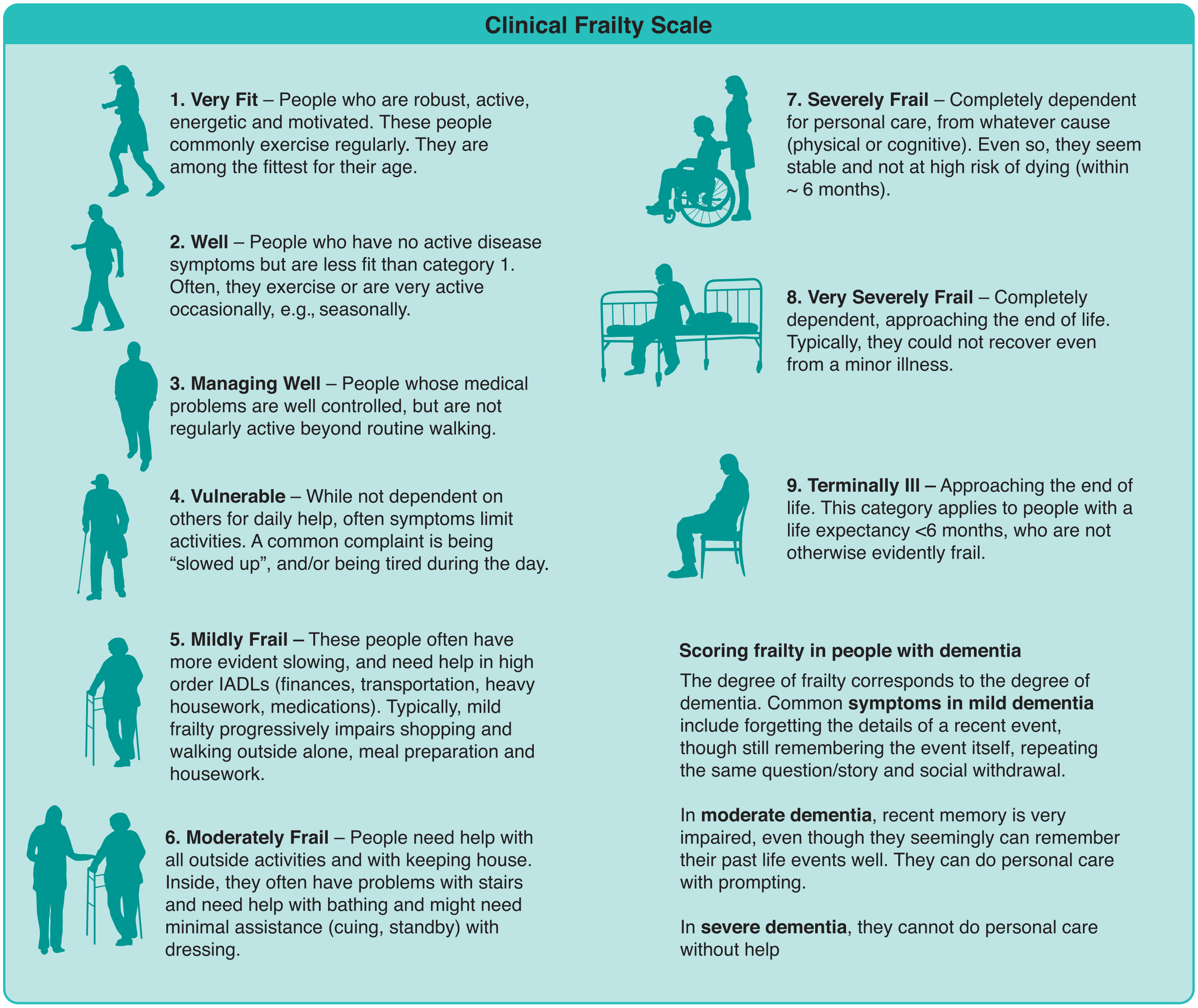

Fig. 84.4 Clinical frailty scale. *IADLs,* Instrumental activities of daily living. (From Rockwood K, Song X, MacKnight C, et al. A global clinical measure of fitness and frailty in elderly people. *CMAJ.* 2005;173(5):489–495.)

Recent estimates have shown that individuals with CKD stage 3b have a threefold increased risk of developing dependence in daily activities such as bathing, dressing, and personal care compared with individuals without renal impairment.[78] Although causation cannot be proven, several studies have suggested that dialysis initiation may exacerbate or accelerate functional decline (Fig. 84.5).[79–81] Most at risk are those with multiple comorbidities, those of older age, and those with a functional loss at the time of dialysis initiation. In a representative, cross-sectional sample of patients on maintenance hemodialysis (HD), followed by the Dialysis Outcomes and Practice Patterns Study (DOPPS), 77% of noninstitutionalized patients 70 years or older reported dependency[82] in contrast to 15% of older adults (≥65 years) in the Cardiovascular Health Study.[83]

In the general geriatric literature, functional dependence is recognized as a contributor to subsequent disability, recurrent hospitalization, and increased mortality.[84–89] Several studies have shown an association between poor mobility status (i.e.,requiring assistance with or total dependency for transfers and ambulation) and mortality risk among incident dialysis patients (Table 84.2).[90–96] Functional dependency (increased need for nursing care, less participation in social societal roles) is associated with a twofold increase in mortality (HR, 2.4; 95% CI, 1.9–2.9) for those with the highest degree of functional loss.[82]

Many factors contribute to the high rate of functional dependencies among older patients with advanced CKD. High rates of depression, cognitive impairment, poor physical functioning, multimorbidity, and the need for repeated hospitalizations may all contribute to the patient’s reduced overall health and self-care ability. Early identification of functional impairment in CKD patients may provide an opportunity to ensure adequate management of modifiable contributing factors, provide support and referral to geriatric rehabilitation services, and trigger discussions, with the family

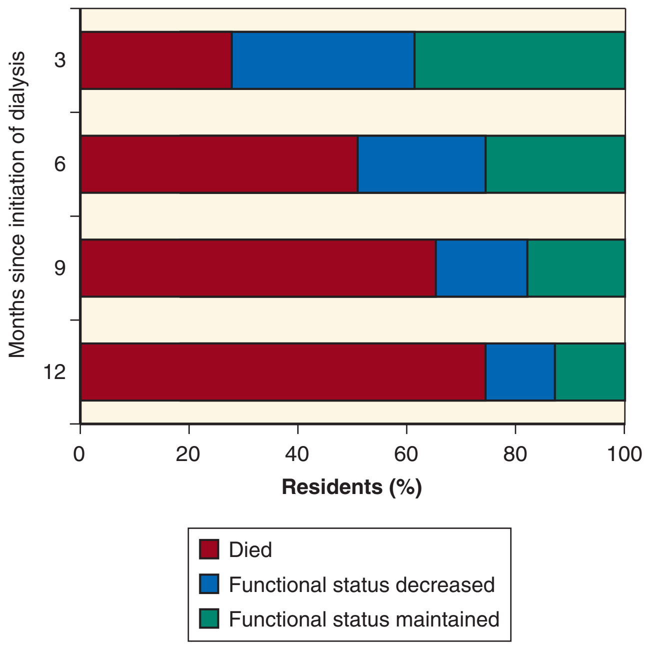

Fig. 84.5 Survival and functional status after the initiation of dialysis among nursing home residents. (From Kurella Tamura M, Covinsky KE, Chertow GM, et al: Functional status of elderly adults before and after initiation of dialysis. *N Engl J Med.* 2009;361:1539–1547.)

and patient, around the prognosis and identification of the trajectory of functional decline in older patients starting dialysis with limitations.

FALLS

As noted previously, several GSs tend to occur concomitantly. Accidental falls are more common in individuals with other GSs, such as concomitant frailty and functional decline, although this relationship is not absolute.[97] Falls are common among older adults and are a leading cause of injury-related hospitalizations, often with prolonged hospital stays, functional decline, long-term care admission, and mortality.[98–102] Up to one-third of community-dwelling individuals older than 65 years fall annually, often recurrently.[103] Injurious falls, such as those causing fractures, head injuries, or cuts, occur in one out of five falls,[104,105] although usually one fall will trigger a cycle of fear of falling (>25% of cases) and a reduction in activity levels, furthering the risk of subsequent falls and the ability to care for oneself.[106]

Falls are almost twofold more common in the dialysis population, both in those maintained on HD and those on peritoneal dialysis (PD).[107,108] Although there are multiple common risk factors for falls (Table 84.3), one of the less appreciated complications of HD that may contribute to falls is the effect of forced immobility. The long time during which patients are sitting relatively still in the dialysis chair may lead to changes in balance, lack of sensory stimulation, and, over time, muscle atrophy. More concerning, however, is the 1.6-fold higher risk of serious fall injuries (i.e., resulting in fractures, head injury, or joint dislocation) that occurs in the year after dialysis initiation, compared with the year prior to dialysis care.[109] Clinical data also support the suggestion that dialysis itself may be causative. In a small single-center study of community-dwelling seniors, falls were more common on dialysis days than on nondialysis days (1.45 vs. 1.35 falls/person, respectively) and, if occurring on dialysis days, threefold more common after dialysis than before dialysis.[102] The association between falls and earlier stages of kidney disease has been less well studied, but some authors have suggested that a reduction in eGFR may be a marker of higher fall risk and that elevated ACR may be an independent predisposing factor associated with serious fall injuries, such as fractures.[98]

Routine clinical questioning about previous falls and conducting simple tests for gait and balance may be a key strategy to identify at-risk individuals who warrant a more detailed evaluation and referral for a CGA. Multidisciplinary risk assessment, followed by tailored multifactorial interventions, such as exercise, treatment of pain, assistive devices, medication review, and home hazard modifications, may be effective in reducing fall risk.

DEPRESSION

Depressive symptoms and clinical depression are common in patients across the complete spectrum of CKD. The prevalence of depression varies, depending on the assessment tools used and the population studied, but, using self-report or screening questionnaires, it is estimated that close to 40% of those undergoing maintenance dialysis have some depressive symptomatology, even if they do not fulfill all criteria for a major depressive episode. Prevalence estimates of major depressive episodes, using formal psychiatrist interview assessments, are lower, ranging from 21% in earlier stages of CKD to 23% in those undergoing maintenance dialysis.[110] These estimates may, however, be influenced by the observation, from a number of studies, that patients on dialysis are often unwilling to undergo formal psychiatric evaluation or treatment. Age-specific data are not available, but depression is often more common in socially isolated individuals and those with functional or cognitive impairment placing frail older individuals at increased risk. In addition often co-occurring with other GSs, depression can be associated with a high burden of pain, fatigue, poor sleep, pruritus, and nausea.[111]

The recognition of depression in the older individual is important for several reasons. In addition to the well-recognized impact on QOL,[112–115] individuals who are maintained on dialysis and are older than 75 years are among those at highest risk of suicide.[116] Furthermore, depression is associated with increased hospitalization and mortality in both CKD and ESKD.[117–122]

The approach to treating depression in older patients is the same as for younger patients. The mainstay treatments are depression-specific psychotherapies, such as cognitive-behavioral therapy (CBT) or interpersonal therapy, and pharmacologic treatment with an antidepressant.

Several studies have shown CBT to be successful in patients with ESKD[123–125] and, where possible, it should be considered the first-line treatment for patients with mild to moderate depression.[126] However, little is known about the influence of physical diseases, frailty, and cognitive impairment on the efficacy or feasibility of psychotherapy. Despite the high prevalence and clinical implications of depression, many patients remain untreated. Several studies have shown low

Table 84.2 Association of Functional Impairment at Time of Dialysis Initiation With Adverse Health Outcomes

Study, Date of Study	Patients	Study Setting	Functional Assessment	Outcome	Association
Arai et al.,[91] 2014	*N* = 202 Mean age, 80 yr	Single-center study, incident to dialysis Age ≥ 75 yr	Mobility (ability or lack of ability to walk without assistance)	6-mo mortality	Higher mortality if unable to walk without assistance; 6-mo mortality risk: aHR, 4.9; 95% CI, 1.4–17.1
Couchoud et al.,[92] 2009	*N* = 2500 Mean age, 81 yr	French REIN Registry data, incident to dialysis Age ≥ 75 yr	Mobility (walk without help, assistance with or total dependency for transfers)	6-mo mortality	Higher mortality if fully dependent for transfers; 6-mo mortality risk: aOR, 1.7; 95% CI, 1.4–2.0
Couchoud et al.,[93] 2015	*N* = 12,500 Age (median), 81 yr	French REIN Registry data, incident to dialysis Age ≥ 75 yr	Mobility (walk without help, assistance with or total dependency for transfers)	3-mo mortality	Higher mortality if requiring assistance for transfers; 3-mo mortality risk: aOR, 2.5; 95% CI, 2.1–2.9 Higher mortality if totally dependent for transfers; 3-mo mortality risk: aOR, 6.5; 95% CI, 5.4–7.9
Dusseux et al.,[90] 2015	*N* = 8955 Age (median), 78 yr	French REIN Registry data, incident to dialysis Age ≥ 70 yr	Mobility (walk without help, assistance with or total dependency for transfers)	3-yr mortality	Higher mortality if requiring assistance for transfers; 3-yr mortality risk: aOR, 1.7; 95% CI, 1.5–1.9 Higher mortality if total dependency for transfers; 3-yr mortality risk: aOR, 3.0; 95% CI, 2.3–3.8
Glaudet et al.,[94] 2013	*N* = 557 Mean age, 71 yr	French REIN Registry data, incident to dialysis	Mobility (walk without help, assistance with or total dependency for transfers)	4-yr mortality	Higher mortality if requiring assistance for transfers; 4-yr mortality risk: aHR, 1.4; 95% CI, 1.2–1.7 Higher mortality if total dependency for transfers; 4-yr mortality risk: aHR, 1.4; 95% CI, 1.0–2.0
Kurella et al.,[96] 2007	*N* = 83,996 Mean age, 84 yr	USRDS Registry data, incident to dialysis Age ≥ 80 yr	Mobility (inability to walk or transfer)	Mortality	Higher mortality if unable to walk or transfer Overall mortality risk: aRR, 1.5; 95% CI, 1.5– .6
Thamer et al.,[95] 2015	*N* = 52,796 Mean age, 77 yr	USRDS Registry data, incident to dialysis Age ≥ 67 yr	Assistance with daily living (requiring assistance with daily living, inability to ambulate, or has an amputation)	3-mo mortality	Higher mortality if requiring assistance with daily living or for walking; 3-mo mortality risk: aOR, 1.4; 95% CI, 1.3–1.5

aHR, Adjusted hazard ratio; *aRR,* adjusted relative risk; *aOR,* adjusted odds ratio.

treatment acceptance and, on average, only one-third of patients who may benefit actually receive antidepressant medication.[127–131]

Pharmacologic therapies have been used in patients with CKD or ESKD; however, there is little information on safety, efficacy, and optimal dosing.[132,133] Most antidepressant medications are highly protein-bound and are not removed by dialysis. Although they commonly undergo hepatic metabolism, the active metabolites are excreted via the kidney, and consequently dosing should be adjusted to the eGFR level (Table 84.4). Based on current data, selective serotonin reuptake inhibitors (SSRIs), such as sertraline, escitalopram, and paroxetine, should be considered first-line for pharmacologic treatment for depression in patients with kidney disease.[134,135] Tricyclic antidepressants (TCAs) and monoamine oxidase inhibitors (MAOIs) should be avoided in older patients with (or without) kidney disease because of cardiac and central nervous system side effects, including prolonged QTc, arrhythmia, anticholinergic effects, and orthostatic hypotension. Paroxetine may be useful when treating a depressed patient with concomitant intractable pruritus because of its antipruritic actions; drugs such as duloxetine may be prescribed, off-label for those with concomitant pain. Treatment doses of duloxetine need to be adjusted and, based on data from a small pharmacokinetic study in HD patients,[136] ideally given at 48-hour intervals. It is important to note that in many regions of the world, this is an off-label prescription, with current US product labeling recommending that the drug not be administered if creatinine clearance (CrCl) is < 30 mL/min. Mirtazapine, a noradrenergic and specific

Table 84.3 Common Risk Factors for Falls in Older Patients With Chronic Kidney Disease

Risks	Details	References
Previous falls	One or more falls during the previous 12 mo	99, 102, 308
Fear of falling	Fearful anticipation of a fall; detectable on screening using the Falls Efficacy Scale (high scores)	106
Frailty	Predisposes to falls for multiple reasons (see text)	108, 309
Balance and gait problems	Increased postural sway, failed walking test; can be exacerbated by arthritic changes or neuropathy	310
Pain	Pain may lead to guarding and abnormal posture due particularly to lower limb and foot pain	
Depression and antidepressants	Fall risk worsened by insomnia and sleep fragmentation, impaired attention and psychomotor slowing, and/or exacerbated by the use of sedatives and antidepressants	308, 310
Number of prescribed oral drugs	Increased fall risk, particularly with benzodiazepines, antidepressants, beta-blockers, other antihypertensive agents, opiate derivatives	310-312
Cognitive impairment	Falls precipitated by decreased awareness of environmental risks; altered judgment and reduced executive functioning result in less effective compensation for age- or disease-associated changes in gait and balance	313
Sensory deficits	Poor vision and/or hearing affect environmental awareness, balance, and compensation strategies	
Diabetes	Impaired vision due to retinopathy, peripheral neuropathy and hypoglycemic episodes increase the risk of falls	310, 314
Anemia	May result in low energy, poor exercise tolerance, and postural hypotension	
Protein energy wasting and sarcopenia	Loss of muscle mass and strength result in less effective strategies to compensate and prevent falling; reduced exercise capacity	308, 315
HD therapy	Increased risk of transient or rapid fluid and electrolyte shifts, hypotension, arrhythmias; stiffness and muscle weakness exacerbated by forced mobility	316
PD therapy	Environmental risks—connected to a cycler machine overnight, tubing around the bed area may pose trip risk; mechanical factors—changes in the center of gravity due to large volumes of intraabdominal fluid	107

HD, Hemodialysis; *PD,* peritoneal dialysis.

serotonergic antidepressant medication, has hypnotic, appetite stimulant, and antipruritic effects and can be used at reduced doses in older renal patients with concomitant anorexia, sleep disturbance, and/or pruritus.

POLYPHARMACY

Polypharmacy is often defined as a medication count of five or more medications.[137,138] Because patients with CKD take on average 8 medications, and those on dialysis take between 10 and 12, polypharmacy is possibly one of the largest issues in renal medicine. In older populations, polypharmacy is associated with increased mortality, adverse drug events and drug interactions, falls and other GSs, low medication adherence, and greater health care costs.[139] Safe medication prescribing in older adults with CKD is complicated not only by the number of medications and presence of comorbidities, but also by the altered pharmacokinetics and pharmacodynamics in the setting of advancing age and presence of CKD. The combination of age-related changes and CKD places older patients at increased risk of drug accumulation and higher potential for adverse drug effects.[140,141] Evidence to guide prescribing is severely limited because a large proportion of clinical trials excluded those older than 65 years, and therefore treatment decisions were often based on evidence extrapolated from other patient groups.[142–144] As a general guide, clinicians are advised to consider multiple factors listed in Table 84.5. Drugs that have a narrow therapeutic index and/or high toxicity levels should only be used when other alternatives are not available, and individuals should be monitored carefully.[145]

Several tools can identify medications in which the risks of use in older adults often outweigh the benefits or identify medications that have been omitted but are likely to benefit the individual. The most widely used tools include the Beers Criteria,[146] Screening Tool to Alert doctors to the Right Treatment (START),[147] and Screening Tool of Older Person's potentially inappropriate Prescriptions (STOPP).[148] START focuses on ensuring that medications that are indicated, and likely to provide benefit, are not omitted in error, whereas the Beers Criteria and STOPP focus more on minimizing the exposure to medications for which the risks outweigh potential benefits. The 2015 Beers Criteria are divided into five sections—medications to avoid in most older adults, medications to avoid in older adults with specific diseases or syndromes, medications to be used with caution in older adults, potentially clinically important drug-drug interactions, and medications to avoid or adjust dose in older adults based on kidney function. The combined STOPP-START criteria, recently updated,[149] have been shown to improve medication appropriateness significantly and reduce adverse outcomes from potentially inappropriate medications.[150] Incorporating the Beers criteria and STOPP-START criteria into a daily drug prescription review process can maximize the efficiency and safety of pharmacotherapy and minimize the number of errors in the administration of drugs to older patients with CKD.

Table 84.4 Safety, Dosing, and Efficacy of Common Antidepressant Medications Used in Chronic Kidney Disease (CKD) and End-Stage Kidney Disease

Drug Class, Generic	Dose Adjustment in CKD[132]	Comment	Efficacy Studies	Class Adverse Effects[132,133,317]
Selective Serotonin Reuptake Inhibitor (SSRI)				
Sertraline	CKD stages 1–4: no dosage adjustment required; 50–200 mg/day CKD stages 5–5D: start at 25 mg/day, consider decreasing max dose	Less than 1% excreted unchanged in urine Pharmacokinetics in CKD are unchanged in single-dose studies, but no published data on multiple dosing Likely not dialyzed	Three prospective efficacy studies[129,318,319]: sertraline significantly improved BDI scores	Common: insomnia, restlessness, nausea, headache, GI upset, sexual dysfunction, activation (mainly with fluoxetine and sertraline) Rare: SIADH, increased bleeding risk, extrapyramidal symptoms, serotonin syndrome (in combination with other serotonergic drugs), and QT prolongation (seen with dose > 40 mg of citalopram)
Paroxetine	IR: 10–40 mg/day CR: 12.5–50 mg/day	<2% excreted in urine Increased plasma concentration found when GFR < 30 mL/min If GFR < 30 mL/min, start at 10–20 mg/day, increase slowly Likely not dialyzed	One study among HD patients ($N = 34$)[320]: paroxetine significantly improved depressive symptoms and nutritional markers	
Citalopram	No dosage adjustment required: 10–40 mg/day	<15% excreted in urine Manufacturer does not recommend use if GFR < 20 mL/min Dose adjustment normally not required in renal impairment, but use with caution when GFR < 10 mL/min Only 1% of drug removed by HD	One randomized study (citalopram versus psychological training) in HD population ($N = 44$)[321]: both citalopram and psychological training significantly reduced HADS scores at the end of 3 mo	
Escitalopram	No dosage adjustment required: 10–20 mg/day	<10% excreted in urine Manufacturer does not recommend use if GFR < 20 mL/min Likely not dialyzed	One randomized study[322] (escitalopram vs. placebo) in 58 ESKD patients—escitalopram significantly improved HADS scores compared with placebo	
Fluoxetine	No dosage adjustment required: 20–60 mg/day	5%–10% excreted in urine Long half-life. If GFR < 20 mL/min, consider using on alternate days or low dose Likely not dialyzed	Two small prospective efficacy studies—fluoxetine improved depressive symptoms by more than 25% in six HD patients[323] and, compared with placebo, improved depression at 4 wk but not 8 wk in 14 HD patients[324]	
Serotonin-Norepinephrine Reuptake Inhibitor (SNRI)				
Venlafaxine	Normal dosage: 75–225 mg/day eGFR 10–70: consider reducing dose by 25%–50%. Dialysis: reduce total daily dose by 50%	5% excreted in urine, clearance decrease and half-life prolonged in renal impairment Manufacturer advises to avoid use if GFR < 10 mL/min Accumulation of toxic metabolite can occur, rhabdomyolysis and renal failure have been reported, but rare	No efficacy data	Similar to SSRIs plus increased blood pressure Liver toxicity seen with duloxetine
Duloxetine	No adjustment required if eGFR > 30: 40–120 mg/day; use not recommended with eGFR < 30	Off-label use in patients with advanced CKD and on maintenance dialysis	No efficacy data	

Continued on following page

Table 84.4 Safety, Dosing, and Efficacy of Common Antidepressant Medications Used in Chronic Kidney Disease (CKD) and End-Stage Kidney Disease (Cont'd)

Drug Class, Generic	Dose Adjustment in CKD[132]	Comment	Efficacy Studies	Class Adverse Effects[132,133,317]
Noradrenergic and Specific Serotonergic Antidepressant				
Mirtazapine	No dosage adjustment recommended for eGFR 10–50: 15–45 mg/day GFR < 10: consider dose reduction	75% excreted unchanged in urine Clearance reduced by 50% in those with GFR <10 mL/min Has been used to treat pruritus caused by renal failure	No efficacy data	Appetite stimulation, weight gain, sedation
Norepinephrine-Dopamine Reuptake Inhibitor				
Bupropion	Consider reduced dose and/or frequency: 150–450 mg/day		No efficacy data	Increased risk of seizures, insomnia, anxiety, decreased appetite
Tricyclic Antidepressant (TCA)				
Nortriptyline	No dosage adjustment required: 35–150 mg/day Not recommended for older adults, especially those with CKD	<5% excreted in urine All metabolites are highly lipophilic.	No efficacy data	Anticholinergic effects, orthostatic hypotension, sedation, cardiotoxicity
Amitriptyline	No dosage adjustment required: 10–200 mg/day Not recommended for older adults, especially those with CKD	<2% excreted in urine	No efficacy data	

BDI, Beck Depression Inventory; *CR,* controlled release; *HADS,* Hospital Anxiety and Depression Scale; *IR,* immediate release; *SIADH,* syndrome of inappropriate antidiuretic hormone.

MALNUTRITION AND PROTEIN-ENERGY WASTING

Depending on the definitions used, anywhere between 25% and 75% of older patients maintained on dialysis have evidence of uremic malnutrition, muscle wasting, and body fat loss.[151] In the renal literature, the term "protein-energy wasting" (PEW) is used in preference to malnutrition because CKD-related factors contribute to the development of wasting, independent of the nutrient intake.[152] A full review of PEW is beyond the scope of this chapter; however, it is important to recognize PEW in the older patient because it is associated with weakness, falls, and increased mortality and remains largely unrecognized and undertreated.[153,154]

Appetite loss is relatively common in older patients with ESKD, with an estimated one-third to one-half of patients experiencing a progressive, spontaneous decrease in food intake as CKD progresses.[155,156] This change in appetite often drives the initial changes in nutrient intake[152]; however, the spiral of decline in overall nutrition is compounded by multiple factors common among older individuals (Fig. 84.6). In renal medicine, the introduction of dietary restrictions affects variety, flavor, and palatability of food and further increases the undernutrition. In many cases, the loss of the social pleasures of eating leads to a low QOL, in addition to PEW.

Special attention should be given to nutritional status in older individuals, and personalized support from a renal dietitian is imperative when PEW is noted—both the subjective global assessment (SGA) and malnutrition inflammation score (MIS) are reliable screening tools. Interventions to improve PEW are often limited to supplementation, with little evidence to support changes in dialytic interventions. Symptoms of constipation and/or dry mouth should be aggressively treated and, where possible, the use of drugs known to reduce appetite should be avoided. Although not evidence-based, mirtazapine and/or other pharmacologic stimulants, together with relaxation of all dietary restrictions, have been tried.

COGNITIVE IMPAIRMENT

Cognitive impairment affects patients negatively by contributing to functional dependence and behavioral symptoms, which in turn have important effects on patient well-being and QOL. Impaired cognition potentially affects multiple areas of patient care, including patient adherence with treatment plans and medication, and may lead to premature institutionalization and death.[157–159] Pathophysiology, prevalence, and treatments of neurologic conditions in kidney disease are discussed elsewhere in this text (see Chapter 57). Screening is important for all individuals, and relatively quick

Table 84.5 Key Elements in Considering Risks and Benefits When Prescribing for Older Adults With Kidney Disease

Risk Considerations	Benefit Considerations
Medication-Associated Risks	
• Is the medication cleared in whole or in part by the kidney? Does the medication have a narrow therapeutic window? • Is the medication thought to be of high risk in the general older population or in individuals with comorbidities similar to those of the patient in question? • Does the medication have potential central nervous system effects? • Are data available to guide dosing in patients with kidney disease (e.g., pharmacokinetic studies, drug level monitoring)?	• What is the population in which this medication has been studied? Does it include older adults or those with kidney disease? Is there observational or clinical trial evidence that benefits extend to those with kidney disease? • If patients with kidney disease have not been studied, does the medication have a track record of safety in postapproval studies? • Would benefits of the medication be due to improvement in symptoms or in decrease in risk from asymptomatic disease? What are the patient's preferences in adding new medications? • Does the medication address a problem for which the patient is at significant risk (e.g., cardiovascular disease in patients with kidney disease)? Is the patient likely to accrue significant absolute risk reduction from the new medication?
Patient-Associated Risks	
• Is the patient already taking multiple medications? • Does the patient have cognitive dysfunction, poor vision, frailty? Risk of additional medications may be higher in this group. • Does the patient have a history of adherence problems, and what would be the consequences of patient-related erratic dosing?	

From Rifkin DE, Winkelmayer WC: Medication issues in older individuals with CKD. *Adv Chronic Kidney Dis.* 2010;17:320–328.

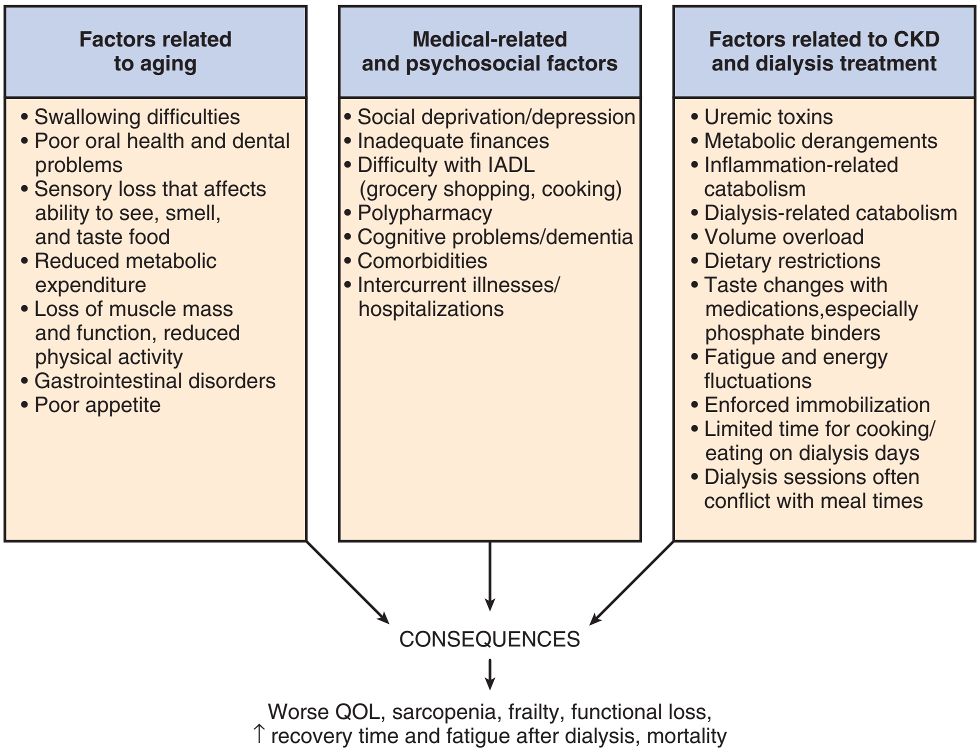

Fig. 84.6 Common factors related to aging and dialysis leading to malnutrition and protein-energy wasting (PEW). *CKD,* Chronic kidney disease; *IADL,* instrumental activities of daily living; *QOL,* quality of life.

tests such as the Mini-Cog (three-word recall and clock drawing) or the Mini-Mental State Exam (MMSE) may be useful in a busy practice. Because cognitive impairment deteriorates over time, particularly after dialysis initiation,[160] it is important to recognize and discuss any features of cognitive impairment that are present and introduce the patient and family to the high likelihood of further progression to dementia. Implementation of adequate social support and planning for future patient care is a critical part of these discussions.

CONTROVERSIES AND COMPLEXITY IN THE MANAGEMENT OF OLDER PATIENTS WITH CHRONIC KIDNEY DISEASE

Over the last decade, multiple clinical practice guidelines have emerged, creating an evidence-based structure that is currently used to guide clinical CKD care. Clinical practice guidelines help standardize care and ensure a consistent approach, across all patients, that includes the interventions most likely to reduce mortality and cardiovascular events and to slow disease progression. Guidelines incorporate data from large population studies and consequently their strength—the ability to create a simplified model to guide consistent high-quality care—is also, in some respects, their greatest weakness. Individual patients bring their own complexities, values, and goals and, particularly when applied to frail older individuals approaching the end of life, guidelines may lead to care that is poorly aligned with the individual's needs. Clinicians are encouraged to solicit individual patient goals and values, distinguish between younger and older individuals, robust or frail, and between those with renal-limited CKD and those with complex clustering of multiple chronic diseases.

Within the renal sphere, guidelines have given rise to several discussions around whether adaptation is required when treating an older frail patient. These controversial issues are introduced in the subsequent sections.

HYPERTENSION IN THE FRAIL OLDER PATIENT WITH CHRONIC KIDNEY DISEASE: TARGETING BLOOD PRESSURE

Decades of research have unequivocally shown that antihypertensive treatment reduces the risk of cardiovascular (CV) outcomes in several populations, including older and very old patients. Currently, however, there is uncertainty about how strict blood pressure (BP) control should be in older individuals with advanced kidney disease.[161,162] Although a discussion of all hypertensive trials is beyond the scope of this chapter and is well covered elsewhere (see Chapter 49), data pertinent specifically to older individuals with CKD are included later.

Newer guidelines, such as those in Canada and Australia, have incorporated data from the Systolic Blood Pressure Intervention Trial (SPRINT)[163] published in 2015 promoting systolic BP targets < 120 mm Hg.[164–166] SPRINT is a prospective randomized controlled trial, funded by the National Institutes of Health, that has randomized 9361 patients with high CV risk to standard treatment targeting systolic BP < 140 mm Hg or to intensive BP-lowering targeting systolic BP < 120 mm Hg. SPRINT included adults aged 50 years or older who were at modest risk of vascular disease but who were free of diabetes mellitus and did not have a recent stroke, CV event, or heart failure. SPRINT showed that in this select population, lowering systolic BP to a goal of <120 mm Hg resulted in significantly lower rates of fatal and nonfatal CV events and death from any cause. The benefits of stricter BP goals were also confirmed in a prespecified subgroup analysis of 2636 subjects aged 75 years or older.[167] Of interest, although the overall rate of serious adverse events was not different between groups (38.3% in intensive vs. 37.1% in the control group; $P = .25$), more patients in the intensive arm developed AKI, syncope, electrolyte abnormalities, and hypotension (but had no injurious falls).

However, there is concern that the data from SPRINT may not be applicable to the older, frail patient population with advanced CKD.[161,162,168] Although a proportion of those aged 75 years or older recruited in SPRINT had elevated creatinine levels (16% had an eGFR < 45 mL/min/1.73 m^2), few had other features of progressive kidney disease (see earlier discussion on controversies around defining older adults as having a disease based on eGFR thresholds alone).[167] Patients with proteinuria > 1 g/day and those with polycystic kidney disease were actively excluded. Unlike many CKD patients, BP was easily controlled with a minimal number of medications. Furthermore, patients with dementia, unintentional weight loss > 10%, or residence in a nursing home were excluded, as were individuals with medical conditions limiting life expectancy to less than 3 years (including cancer).

The benefits of treating hypertension have been confirmed in a recent meta-analysis incorporating data from almost 45,000 patients across all ages.[169] The evidence for which BP targets to use is less clear. Weiss et al.[161] have shown robust evidence that BP targets of less than 150/90 mm Hg led to a reduction in mortality, cardiac events, and stroke but only low to moderate evidence that a BP lower than 140/85 mm Hg was beneficial in adults aged 60 years or older. In addition, they found lower BP targets to be associated with an increase in syncope and medication burden but not higher rates of falls or cognitive decline. Although the rate of adverse effects seemed similar across all BP targets, the impact of an adverse event is likely higher in older patients, particularly those residing in a nursing home,[170] and consequently caution is advised. At least, at present, as data continue to emerge, the reader is best advised to integrate the results of a CGA into decision making and adapt treatments to align with individual patient complexity.

THE ROLE OF STATINS IN OCTOGENARIANS

CKD is associated with an increased risk of major CV events and mortality and is sometimes considered a coronary risk equivalent to diabetes and hypertension.[171] Although dyslipidemia is an important modifiable risk factor for CV disease in the general population, particularly in those older than 50 years, its role and relevance in the older CKD population are less clear. Current KDIGO guidelines recommend the initiation of statin therapy, either alone or in combination with ezetimibe, for all nondialysis patients aged 50 years or older with an eGFR < 60 mL/min/1.73 m^2, irrespective of age or baseline lipid profile.[172] Much of the evidence is based on high quality data from the Study of Heart and Renal Protection (SHARP) discussed later.[173] Unlike contemporary guidelines for the general population, published by the American College of Cardiology and American Heart Association,[174] that apply only to those aged 20 to 79 years, there is, however, no upper age limit included in the KDIGO guidelines. As a result, it may be appropriate to take a more cautious approach, particularly in those with multimorbidity, using guiding principles published by the American Geriatric Society.[175]

The SHARP trial, published in 2011, randomized patients with CKD to 20 mg simvastatin plus 10 mg ezetimibe or placebo.[173] Patients were followed prospectively for a median of 5 years, and results showed a lower incidence of cardiac death, stroke, nonfatal myocardial infarction, and cardiac revascularization (11.3% vs. 13.4%, respectively). A number of participants were aged 70 years and older (28% of SHARP participants), and the effectiveness of the interventions, simvastatin and ezetimibe, was similar. There was little evidence from the data presented that suggested any increased risk of adverse events in these older participants. The dilemma, however, arises from what, and who, was not included in the study. A number of patients were excluded because their physician did not think that participation was appropriate. Many of these may have been older individuals with social or other characteristics not captured in the study data. In an eloquent piece, Butler et al. have argued that application of current KDIGO guidelines advocating for the use of statins in those of any age with CKD may not be appropriate.[176] They suggested that the older patients recruited into the study were likely not representative of many of those attending our clinics; they referred specifically to patients with a history of myocardial infarction, cognitive impairment, or functional limitations who would have been excluded, but also allude to those with social or psychological frailty. Also, they remind us of the need to elicit treatment preferences and goals, especially because both pill burden and cost are often more important to older individuals than to younger patients, and the impacts of any adverse drug effects are higher.

Whereas the benefits of lipid lowering have been shown for patients with nondialysis CKD, the benefits do not appear to extend to dialysis patients. Several large randomized trials and high-quality meta-analyses have suggested that statin therapy has little or no effect on cardiovascular outcomes among adults undergoing maintenance dialysis.[173,177–183] Furthermore, emerging evidence has suggested that statins may accelerate vascular calcification in the dialysis population.[184] As a result, the indication for use in older patients, particularly those with frailty or multimorbidity in whom the life expectancy is a few years, is weak. At present, the 2014 KDIGO Lipid Guidelines[172] and 2016 Canadian Cardiovascular Society Guidelines[185] recommend statins not be started in adults receiving dialysis, but that they be continued in patients already receiving them at the time of dialysis initiation. Although not supported by evidence or by guidelines, discontinuation of statins may have a role in older individuals maintained on dialysis who present with increasing frailty, multimorbidity, polypharmacy, or malnutrition and those entering the end of life.

ANTICOAGULATION IN ATRIAL FIBRILLATION

The prevalence of atrial fibrillation (AF) in patients with impaired kidney function is considerably higher than that seen in the general population.[186,187] Adults aged 65 to 74 years, undergoing maintenance dialysis, have a prevalence of AF of 13%. This increases to 23% among those 85 years or older.[188] A large proportion of these patients are treated with oral anticoagulants, specifically warfarin, with the intention to reduce the risk of stroke and systemic embolism.

There is mounting evidence that the benefit-risk ratio of warfarin may not be extrapolated to patients with ESKD. At the center of the controversy is the question if, when used in patients with AF undergoing dialysis, warfarin is less effective at stroke risk reduction and is associated with increased bleeding risk.[189,190] Recent observational studies have shown high rates of early discontinuation (47% stop after 8 months)[191] and physician uncertainty.[192,193] At present, there are conflicting guidelines across the world. Both the 2012 KDIGO[9] and 2016 Canadian Cardiovascular Society Guidelines[194] recommend against anticoagulation in patients with a creatinine clearance < 15 mL/min with AF (including patients on dialysis); the 2014 American College of Cardiology/American Heart Association Practice Guidelines[195] suggest anticoagulation with warfarin to an international normalized ratio (INR) target of 2.0 to 3.0 in patients with nonvalvular AF, with a CHA2DS2-VASc score of 2 or higher. The 2012 American College of Chest Physicians[196] and 2016 European Society of Cardiology[197] have made no specific recommendation for patients with ESKD. For older patients, particularly those with one or more GSs, the risks appear to equal or outweigh benefits[198,199] and, at least until results of ongoing trials are available,[200,201] caution is advised when prescribing warfarin for stroke prevention, especially in frail older patients, those with a history of major bleeding and those with known vascular calcification.

Currently, there is optimism that non–vitamin K antagonist oral anticoagulants (NOACs) will offer an alternative anticoagulant for the dialysis population.[189] Dabigatran, rivaroxaban, apixaban, and edoxaban have been studied in large phase III trials in the setting of nonvalvular AF in the general population and were found to have an overall favorable risk-benefit profile versus warfarin.[202] However, only a limited number of small controlled trials of NOACs have included patients with severe CKD (CrCl < 25–30 mL/min), and little is still known about the influence of HD on their pharmacokinetics. Nevertheless, the use of these drugs has become more prevalent, with an estimated 12% of dialysis patients on anticoagulation being treated with NOACs.[203,204]

ANTIRESORPTIVE TREATMENT: BONE HEALTH

Bone health remains a challenging aspect of renal care, amplified in the older population by the attendant high risk of fragility fractures, adynamic bone disease, and polypharmacy. In this section, we review the data (or lack thereof) supporting the use of antiresorptive therapies in older patients who have recently had a fracture.

Fragility fractures, fractures that occur as a result of a low-energy trauma such as a fall from standing height or lower, are common in older people and are associated with excess mortality, prolonged rehabilitation, long-term pain, and high medical costs. This is particularly the case with hip fractures, which have a substantial impact on an older individual's health, QOL, and abilities, with only 50% of patients regaining their prefracture level of mobility and independence.[205] In 2000, an estimated 1.6 million hip fractures occurred worldwide.[206] This number is expected to increase to 6.3 million by 2050.[207] The risk of fracture increases with CKD progression, resulting in risks 1.5 to 4 times higher in older individuals versus those with normal renal function (Fig. 84.7).[208,209,300] Hip fractures occurring in dialysis patients are particularly concerning because the 1-year mortality after hip fracture in the dialysis population is close to 65% versus 20% in the general US population.[210]

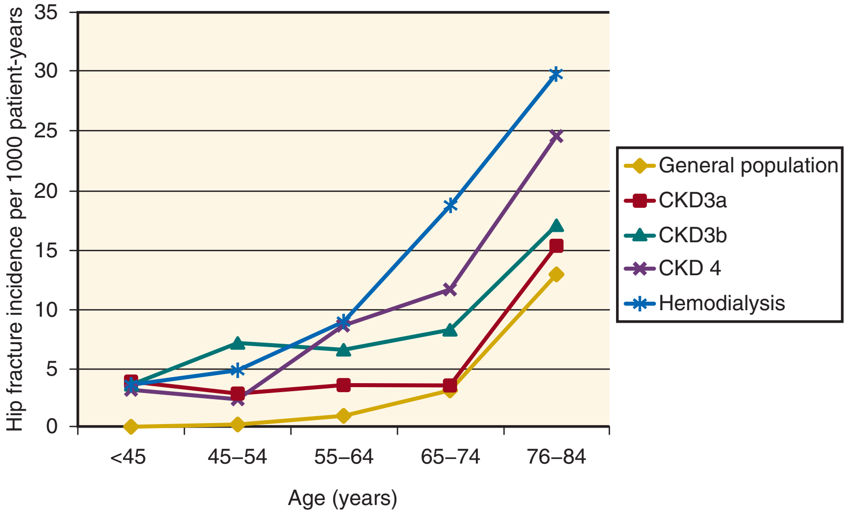

Fig. 84.7 Hip fracture incidence increases with progressive chronic kidney disease (CKD). (From Moe SM, Nickolas TL. Fractures in patients with CKD: time for action. *Clin J Am Soc Nephrol.* 2016;11(11):1929–1931.)

Despite these alarming numbers, there are sparse data to guide clinicians providing postfracture care to older patients with CKD. Older patients with CKD stages 4 to 5D have been uniformly excluded from clinical trials evaluating pharmacotherapies. The uniqueness and complexity of CKD in the pathogenesis of fractures, mixed with age-related and postmenopausal causes of bone loss, render this population very challenging in terms of diagnosis and management of osteoporosis. Active management of hyperparathyroidism and phosphate-lowering therapies may have an effect on fracture risk, but may also negatively affect other predisposing risks such as falls, malnutrition, and aging itself.

There have been few studies regarding the use of bisphosphonates or denosumab in patients with CKD stages 4 to 5D. Clinical experience is limited, and the theoretic risks of administering antiresorptives must be carefully weighed against unknown benefits, particularly because bone pathology can be difficult to determine accurately. Consequently, the widespread use of osteoporosis therapies in patients with advanced CKD and/or with evidence of CKD–mineral and bone disorder (MBD) is discouraged, especially in frail older individuals who, despite an increased risk of recurrent fractures, are likely to suffer equally from adverse effects and potential harm.[211] Antiresorptives and other osteoporosis therapies can be used in patients with modest reductions in eGFR (≥30 mL/min/1.73 m^2) and a parathyroid hormone (PTH) level within the normal range.[212] However, for individuals with severe CKD (stages 4–5D) and those maintained on dialysis, there are limited therapeutic interventions that are known to reduce fracture recurrence. Standard therapies for osteoporosis may be ineffective or possibly harmful in this population. For example, antiresorptives may worsen low bone turnover, osteomalacia, and mixed uremic osteodystrophy and exacerbate hyperparathyroidism; denosumab may induce significant hypocalcemia.[213] In the 2017 KDIGO guideline update, treatment recommendations are discretionary and recommended only after "taking into account the magnitude and reversibility of the biochemical abnormalities and the progression of CKD, with consideration of a bone biopsy."[212]

Nephrologists should continue to pay careful attention to emerging information in this relatively new field while prospective studies, specifically designed for CKD patients, are performed.

VASCULAR ACCESS CREATION

Vascular access creation in older patients undergoing hemodialysis has remained a topic of ongoing discussion over several years.[214–217] In the older patient, it is more important to prioritize the choice of dialysis access based on the treatment goals. Although high-quality studies have shown that dialysis through an arteriovenous fistula (AVF) is associated with a lower infection risk and remains optimal for patient survival,[218–220] there is little doubt that in the field of geriatric nephrology, there are several situations in which a fistula may not be the most appropriate approach.

Earlier studies comparing AVFs with arteriovenous grafts (AVGs) or central venous catheters (CVCs) have likely overestimated the survival benefits for those dialyzed via an AVF. By comparing outcomes of patients who underwent an unsuccessful fistula attempt with those who did not have fistula creation attempted, it appears that the observed survival gains are attributable to residual confounding factors, arising from the inherent bias that individuals undergoing surgery are likely healthier than those for whom surgery is not attempted.[221,222] Furthermore, although the relative gains may be similar regardless of age, the absolute gains, in terms of survival days, are likely to be small because overall survival on dialysis is lower in older individuals. DeSilva and colleagues[223] analyzed data from 115,425 incident US patients on hemodialysis aged 67 years and older. They assessed mortality outcomes based on the first vascular access placed but found no difference in mortality for patients with an AVG as the first access compared with those who started with an AVF. Similar results were reported by Drew et al.[224] using decision analytic techniques to combine current knowledge of the risks and benefits of different access strategies. Although their study has faced minor criticism for the assumption that most patients initiate dialysis with a CVC, the methodology used was robust, valid, and widely generalizable across the world. Results have shown that the mean estimated survival benefit for those aged 80 years is modest when patients are dialyzed using a fistula strategy. For example, life expectancy gains were maximal in men and in nondiabetic patients, with the estimated prolongation of life for an 80-year-old nondiabetic male estimated at 8 months when a fistula strategy was preferred over a CVC strategy and 6 months when compared with AVG. Survival for an 80-year-old diabetic female was best with an AVG and, when comparing AVF with CVC, survival differed by only 3 weeks.[224]

Access-related morbidity is also known to be high. Patients with an AVF or an AVG are more likely to be bothered by pain, bleeding, bruising, and swelling than those with a CVC.[225] Cannulation-related complications are more common, in part due to fragile skin and in part related to the higher use of antiplatelet or anticoagulant medications.[226] Access creation is also associated with reduced grip strength and impaired arm and hand function.[227] In patients with marginal functional status, creation of AVFs or AVGs may affect personal independence and limit dressing, bathing, and other activities of daily living. Furthermore, in studies of octogenarians who underwent fistula creation during the predialysis period, one third to one half died before they needed renal replacement therapy (RRT).[223,228] Other studies have shown fistula failure rates to be high in older patients, with some suggesting up to 60% of older adults will have a fistula that fails to mature.[229–231] Of those that do function, several will require interventions, particularly early in the clinical course, causing a high clinical and economic burden when overall survival and gains are limited.[217]

The ideal clinical scenario is when results of the individual's CGA help inform the choice of vascular access.[232] AVGs in predialysis patients placed closer to the time of HD initiation have been shown to be an effective CVC-sparing strategy[233] and to be less costly than placing an AVF.[217] Although we await the results of ongoing randomized controlled trials comparing access approaches in an older patient population,[234] we recommend a patient-first approach that identifies "the right access for the right patient" based on patient preferences, priorities, life expectancy, and QOL.

CARE FOR OLDER ADULTS WITH END-STAGE RENAL DISEASE

DIALYSIS

The demographics of the dialysis population have been changing over time. Compared with the early 2000s, patients are older and more medically complex[235] with more than 50% being older than 65 years and 25% older than 75 years.[236] Until recently, the adjusted ESKD prevalence has also been rising, with steeper increases among older age groups compared with those younger than 65 years. These trends, however, have started to change, and data from USRDS have suggested more conservative dialysis practice patterns, with pronounced declines in the number of older people starting dialysis resulting in 2016 incidence rates being the lowest among those aged 65 years and older since the late 1990s.[236]

Although it is often assumed that treatment with dialysis will extend life and alleviate the signs and symptoms of advanced kidney disease, there is growing evidence that these benefits may not always accrue in older adults. The median life expectancy after the initiation of chronic dialysis in the United States is comparable to (or worse than) many cancers, and is estimated to be less than 2 years for patients aged 75 years or older. However, considerable heterogeneity exists and, although almost 25% of those aged 70 years or older survive less than 6 months, a similar proportion survive for more than 4 years.[79,96] For all ages, the relative median life expectancy after starting dialysis is close to 25% that of age-matched individuals from the general population,[236] and therefore older adults are most likely to have the smallest absolute gain measured in months or years. Among factors that predict early mortality are the presence of frailty, impaired functional status, low body weight or serum albumin concentration, and need for nursing care at the time of dialysis initiation.[90,95,237–241] There are several prognostic tools,[92,93,237,242] but these are limited to use in individuals who initiate dialysis and do not help prognosticate survival in those who are already established on dialysis or who opt for a comprehensive conservative renal care approach. A recent review has highlighted the limits and deficiencies of the most commonly used prognostic tools.[243]

DIALYSIS MODALITY SELECTION

Among patients 65 years of age and older initiating dialysis in the United States, most of them (92%) will start in-center HD as their initial modality choice, followed by PD (7%) and preemptive transplantation (1%),[236] as shown in Fig. 84.8. Until recently, one of the main arguments favoring HD was based on the overall mortality rates seen in the older patient population treated with PD. In a recent meta-analysis

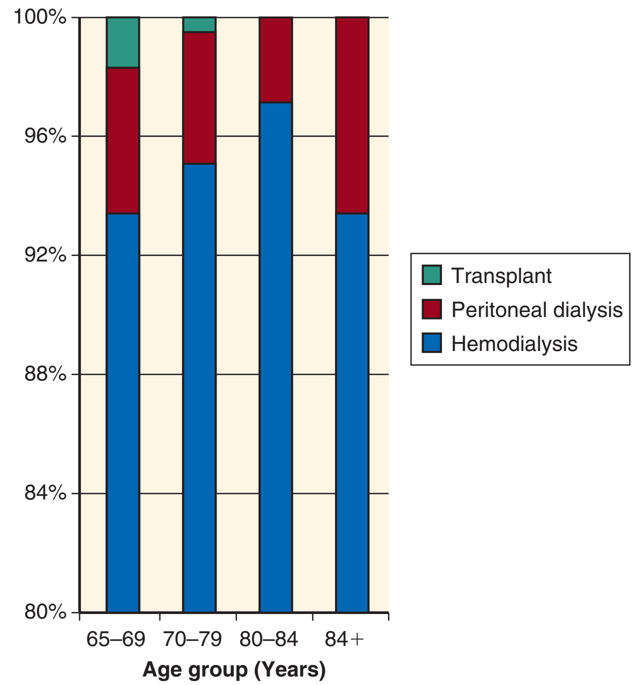

Fig. 84.8 Initial kidney replacement therapy modality in the United States in 2008, according to age group. (From Tamura MK, Tan JC, O'Hare AM: Optimizing renal replacement therapy in older adults: a framework for making individualized decisions. *Kidney Int.* 2012;82:261–269.)

involving 631,421 older patients, the mortality was higher in those undergoing PD compared with HD (HR, 1.10; 95% CI, 1.01–1.20).[244] However, particularly within the field of geriatric nephrology, it is important to recognize that the flexibility and gentle nature of PD offers much to an older individual, and many individuals forfeit a longer time alive but feeling unwell for a shorter time feeling well. Patients may be attracted to several other aspects of PD, such as increased dietary freedom, decreased hospital visits, control over one's own care, and a feeling of personal autonomy.[245] In those with a guarded prognosis but high symptom level, PD may be an ideal alternative to comprehensive conservative renal care because it can be adapted to patients' needs as they transition along their disease course until they reach end of life.[246] For example, in a patient with progressively worsening limb ischemia, who is experiencing a deterioration in his or her health combined with increasing pain and frustration, dialysis exchange frequency can be reduced, and the solution volume and concentration adapted, to improve comfort, prevent shortness of breath, and enhance the feeling of control and freedom. Frail older patients who are unable to perform their own PD exchanges can be successfully supported by assisted PD.[247] With assisted PD, home care workers, family members or staff in long-term care facilities are trained to perform the exchanges and supervise the dialysis. Different models have been used across different countries, and currently neither functional dependence nor cognitive impairment are considered absolute barriers to PD.[248–250]

In-center HD may be preferred over home-based PD treatment by some individuals because their day can be structured around scheduled and fixed times, facilitating coordination for family members or staff providing support. Others benefit from socialization with staff and other patients at the dialysis facility and from having frequent contact with the dialysis staff to provide both medical and psychosocial support.[245,251] HD in frail older patients is, however, more likely to precipitate episodic postural hypotension and possibly accelerate cognitive decline from myocardial and cerebral stunning. Additional challenges include the creation and maintenance of vascular access, problems associated with postdialysis recovery time, such as fatigue, and the risk of falls after dialysis.[252]

As a result, there is no one dialysis modality of choice for the frail older patient. Clinicians are advised to use information from the CGA to build a framework of total care, focusing on the needs of the individual patient and the goals of care in the context of her or his life experience.

KIDNEY TRANSPLANTATION

Patients aged 65 years or older currently comprise almost 25% of all kidney transplant wait-listed patients; 19% of all recipients are older adults.[253] Over the past 2 decades, there has been a fivefold increase in the number of patients aged 65 years and older undergoing kidney transplantation in the United States.[253] Although advanced age is no longer a barrier to kidney transplantation, age remains an important surrogate factor when assessing suitability for transplantation. Chronologic age is less important than the patient's level of frailty, burden of coexisting conditions, and overall prognosis.[254,255] Often, transplantation assessments include more rigorous assessment of cardiac and cancer risk.[256]

Transplantation offers survival benefits across all ages, although due to the attendant risks of the surgery itself and side effects associated with immunosuppressive drugs, patients have an increased mortality risk in the initial few months. Survival equivalence, as compared with maintenance dialysis, is seen approximately 8 months after the surgery is performed in younger patients[257] but not seen in those aged 70 years and older until more than 10 months have elapsed from the date of transplantation.[258] As a result, there is little survival advantage in the initial 18 months posttransplantation.[258] Several strategies have been used to maximize kidney transplantation success, including the use of expanded criteria donor (ECD) kidneys and pairing organs harvested from older donors with older recipients.[259] These systems offer advantages and disadvantages, but overall the most beneficial and commonly used strategy in North America is to offer ECD kidneys to the older patient.[260]

The most common cause of graft loss in older kidney transplant recipients remains death with a functioning graft.[261,262] Death-censored graft failure risk remains lowest, across all age groups, in patients aged 65 years or older likely because of a shorter life span and lower risk of acute rejection. Older patients are at higher risk of medication-related adverse events and higher rates of opportunistic infections and neoplasia.[263,264]

COMPREHENSIVE CONSERVATIVE RENAL CARE

Although the change in dialysis utilization is captured across several regions of the world through dialysis registries, little is known about how often patients with advanced stages of CKD choose not to undergo dialysis or are not offered dialysis. Single-center studies from Europe and Australia have suggested that a substantial number of older patients with advanced kidney disease are treated conservatively and do not receive dialysis.[265–270] Other studies, using administrative data, also found that a notable number of older patients do not undergo dialysis. These studies identified patients with advanced CKD who were likely at risk of dialysis by selecting those who had two or more creatinine levels determined at least 3 months apart that met criteria for a low eGFR. In one study, from Canada, the clinical course of all individuals with creatinine values consistent with CKD were tracked over a 2-year period (Fig. 84.9).[271] In the second study, from the VA, detailed data were extracted from electronic charts data and used to identify whether patients either underwent dialysis or were being prepared for dialysis.[272] In both studies, older adults were found to be less likely to undergo dialysis than their younger counterparts, implying that there may be a relatively large "reservoir" of older adults with high creatinine levels (and, by implication, very low levels of eGFR) who, for a number of reasons, do not get started on dialysis (Fig. 84.10). Similar patterns have been reported for the incidence of hospitalized patients with AKI treated with dialysis.[273] Although provocative, these data cannot distinguish between those who did not meet clinical criteria or a need for dialysis from those who did not get offered or who chose not to undergo dialysis.

Although registry data serve as a valuable source of information about outcomes in older adults with ESKD who are treated with dialysis, less is known about outcomes among those who are not treated with dialysis. What little is known

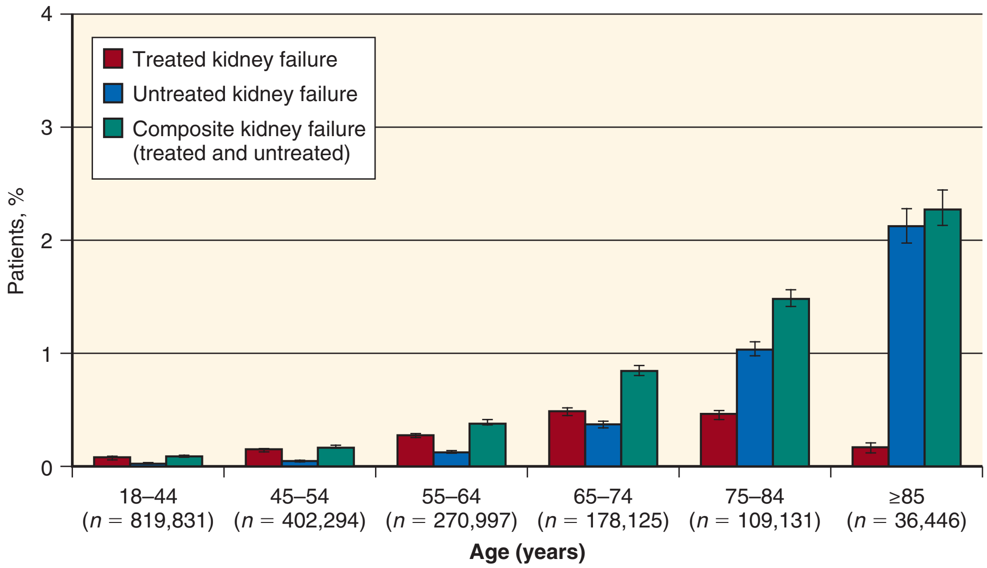

Fig. 84.9 Treated and untreated kidney failure as a function of age in Alberta, Canada. (From Hemmelgarn BR, James MT, Manns BJ, et al: Rates of treated and untreated kidney failure in older vs. younger adults. *JAMA.* 2012;307:2507–2515.)

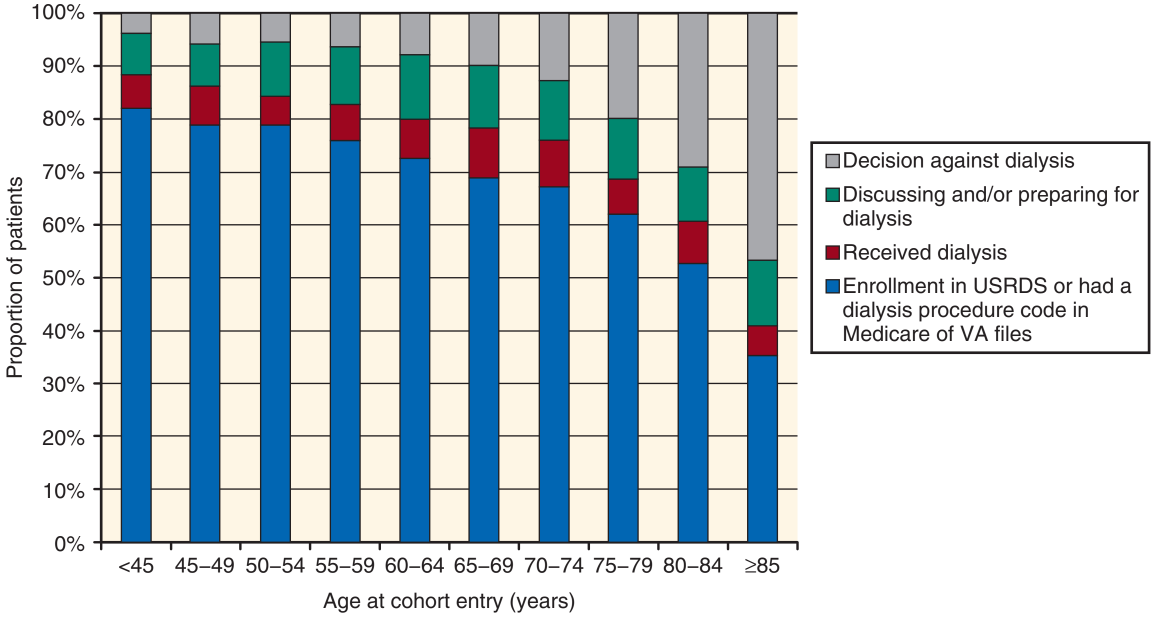

Fig. 84.10 Age differences in treatment decisions and practices for advanced kidney disease. These are based on the basis of a chart review of a random sample of patients who were not enrolled in the US Renal Data System (USRDS) and did not have a dialysis procedure code in Medicare or Department of Veteran Affairs (VA) administrative files during follow-up. (From Wong SP, Hebert PL, Laundry RJ, et al. Decisions about renal replacement therapy in patients with advanced kidney disease in the US Department of Veterans Affairs, 2000-2011. *Clin J Am Soc Nephrol.* 2016;11(10):1825–1833.)

about this group comes from a number of small, single-center studies that have examined outcomes among patients with advanced kidney disease who were managed with comprehensive conservative renal care (CCRC).

Most studies have shown that dialysis is associated with a modest survival advantage over CCRC; however, the magnitude of this effect varies across studies, and the potential advantage seems to be lost for a subgroup of older patients with a high burden of comorbidity.[274–276] The comparisons, however, between patients managed with CCRC and those initiating dialysis are challenging because there is no clear starting point for analysis. It is often difficult to know when dialysis would have been started in those who chose a CCRC pathway, and it is possible that studies suffer from a lead time bias. Robust patients and those with rapid declines in renal function are more likely to undergo dialysis; those who are frail or have slowly progressing renal declines are more likely to undergo CCRC. Thus, the use of time points determined from the eGFR, to set a time when dialysis may have been otherwise started, is likely to lead to further systematic biases.

Evolving literature has recognized that a number of older patients may benefit equally from dialysis as from CCRC. Particularly in older adults with a high burden of geriatric syndromes, small gains in longevity that are potentially achieved with dialysis care need to be balanced against high risk of increased symptom burden, worsening dependency, and often a negative effect on QOL.[252,277,278] The overall gains are modest, particularly in those who have recently undergone one or more life-prolonging procedures, such as intubation, feeding tube placement, and cardiopulmonary resuscitation.[279]

Patients who opt for a CCRC pathway tend to fare well for several months. Their symptom burden and QOL remain stable until the last 1 or 2 months of life, at which time palliative care interventions are often beneficial.[276] Functional status is often maintained until the last months of life and, overall, patients are more likely to die in their home environment or in hospice compared with those starting RRT.[274] Further information on the trajectories seen in those initiating dialysis have been emerging, although insufficient information is currently available to make clear distinctions between patient groups.

IMPORTANCE OF SHARED DECISION MAKING IN END-STAGE RENAL DISEASE

Shared decision making is a critical component of renal care, particularly for older patients and their families. See Chapters 62 and 82, but remember that discussions should be open, collaborative, and structured. Ideally a multidisciplinary renal team, patient, and family should be present and information shared about the potential benefits and harms of the various treatment options. Discussions should be specific to the individual and his or her lifestyle and values. Prognosis should be integrated with the results of the CGA, allowing clinicians to share information about the possible positive and negative effects on daily living and symptom burden. It is important that patients and key individuals in their social network have realistic expectations of life with the different treatment options. Helping patients and families to be part of a problem-solving approach to individualized management will maximize treatment benefits.

DIALYSIS TREATMENTS UNIQUE TO OLDER PATIENTS

DIALYSIS AND GERIATRIC REHABILITATION PROGRAMS

Geriatric rehabilitation programs are designed to optimize the functioning of older adults and often premorbidly frail individuals who have experienced loss of independence due to acute illness or injury, often superimposed on chronic functional and medical problems.[280] Geriatric rehabilitation provides evaluative, diagnostic, and therapeutic interventions to restore functional ability or enhance residual functional capacity in older people with disabling impairments.[281] Physical deterioration occurs in predialysis CKD prior to the commencement of RRT.[282,283] Once patients start RRT, they appear to decrease their levels of physical activity further,[284] leading to increased skeletal muscle wasting and poorer overall physical function.[285,286] This vulnerable state is exacerbated by multiple or recurrent periods of illness or hospitalization, which, even if short-lived, may lead to persistent disability.[287] Over a span of 6 months, vulnerable individuals experience increased dependency in a number of aspects of daily living, which may eventually result in a nursing home placement.[79,80,288] Timely referral for geriatric rehabilitation may reverse recent onset dependence and disability and increase the likelihood that patients return home or to a community residential setting following rehabilitation.[289]

Specialized dialysis rehabilitation programs integrate active rehabilitation with dialysis care. An interdisciplinary team consisting of multiple professionals, including physicians, nurses, pharmacists, physiotherapists, occupational therapists, speech-language pathologists, social workers, psychologists, dietitians, and recreation therapists, work together, often with overlapping roles, to provide interventions such as therapeutic exercises, fall prevention plans, education on ways to conserve energy during activity, and recommendations of assistive devices.[290] Geriatric dialysis rehabilitation often occurs at a turning point in a patient's disease trajectory and can allow a patient to recover personal function after a complicated course in hospital or a recent multisystem illness. It is also an opportune moment to initiate or pursue discussions around goals of care and advance care planning. Providing dialysis in short daily sessions, if feasible, may improve the success of the program. Short daily dialysis (6 times/week for 2 hours) is well tolerated, limits scheduling conflicts and interference between medical and rehabilitation treatments, and may lead to improved nutrition and better participation in therapy sessions. Patients report less fatigue and fewer symptoms associated with rapid fluid shifts.[291]

Limited effectiveness data exist, with the largest information coming from a single-center report of 164 patients. In this series, more than 70% of patients met functional goals and showed clinically meaningful improvements in personal independence.[289,292] Other models of geriatric rehabilitation, including outpatient rehabilitation, community rehabilitation, and fall prevention programs, have not been well studied in the older renal population but have been shown to benefit other clinical geriatric populations and thus warrant further investigation in the CKD population. A practical first step is for nephrology teams to consider if rehabilitation would be beneficial using information gathered through the CGA—for example, at dialysis initiation, after hospitalization or accidental fall(s), or if there is a change in social status (e.g., changing residence or the death of a caregiver or key family member). Nephrologists may collaborate with the rehabilitation team to reevaluate health targets at this time.

PALLIATIVE DIALYSIS

There has been an emerging dialogue around the role of a palliative approach to care for those on maintenance dialysis.[270,293–296] The palliative approach to dialysis care can be defined as a:

> *transition from a conventional disease-oriented focus on dialysis as rehabilitative treatment to an approach prioritizing comfort and alignment with patient preferences and goals of care to improve QOL and reduce symptom burden for maintenance dialysis patients in their final year of life.*[294]

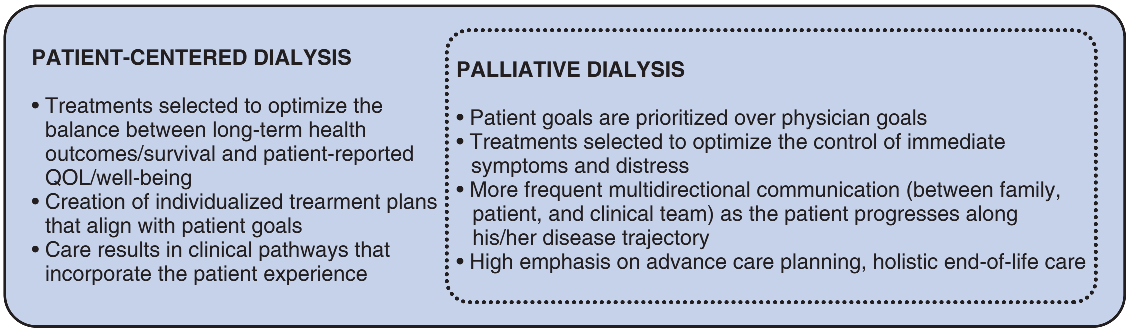

Fig. 84.11 Conceptualizing palliative dialysis. In patient-centered dialysis, treatments are chosen to align with patient goals. In palliative dialysis, a subset within patient-centered care, treatments are chosen so that if needed, patient goals are prioritized over physician goals. *QOL,* Quality of life.

One approach is to classify patients into one of three treatment groups, each with a different expected outcome:

1. Dialysis as a bridging treatment or long-term maintenance treatment, when the patient is expected to return to usual life activities
2. Dialysis as a final destination, when dialysis treatment is used to alleviate symptoms but it is expected that the CKD and/or non-CKD disease will continue to deteriorate
3. Active medical management without dialysis.[297]

Those with high levels of comorbidity, ongoing concomitant nonrenal disease, or an unclear prognosis likely best fit within the second final destination group, and are those who are most likely to benefit from a palliative approach to dialysis care. After shared decision making, dialysis treatments can be customized to alleviate the individual's symptoms and minimize treatment-related distress.[298]

Palliative dialysis is often, incorrectly, thought of as being equivalent to less dialysis or a precursor to dialysis discontinuation. Although it is often necessary to adapt dialysis timings and prescriptions, and dialysis discontinuation may be a component, these interventions are rarely sufficient to ameliorate symptoms and suffering.[293] Ongoing symptom assessment and therapeutic interventions (discussed briefly later) are required. When taking a palliative approach to renal care, advance care planning becomes of paramount importance. As the patient continues along the disease trajectory, he or she, together with the immediate family, will increasingly need support as they learn to accept death as a natural consequence. As death approaches, the focus becomes more consistent with interventions that improve the dying process and end-of-life care. At this point, patients' goals of care tend to shift to focus almost exclusively on QOL and minimizing suffering rather than survival, with a strong emphasis on emotional, social, and family support.[299] Palliative dialysis is a form of patient-centered dialysis in that it continues to align care with patient preferences. In palliative dialysis, the focus of care is on the prevention and relief of suffering. Treatments are adapted to the issues that matter most to the patient and are adjusted so that the individual patient can achieve goals for the remaining life. This approach, also termed "patient-centered care," requires setting a balance between issues important to the clinician (e.g., survival, laboratory targets) with those affecting the individual (restrictions affecting quality of life, symptom control, and physical function; Fig. 84.11).[293] The clinician may therefore need to forgo traditional treatment targets and disease-oriented guidelines in favor of the patient's symptom management and individualized goals of care. This can mean that less emphasis is placed on maximizing long-term health outcomes, such as optimal control of BP, phosphate levels, and other laboratory results, unless they result in immediate symptom improvement.

Common symptoms associated with advanced CKD include fatigue, nausea and vomiting, sleep disorders, restless legs syndrome, depressive symptoms, and pruritus.[293] Many occur in complex clusters and treatments are primarily to ease symptoms sufficiently to maximize comfort because it is not always possible to alleviate them completely.[277,278] Less dialysis rarely provides benefits and may result in increased symptoms and postdialysis fatigue, especially if greater ultrafiltration is required to manage troublesome shortness of breath. However, it can be an option for those in whom fluid management is not a big problem and who favor spending more time at home, with reduced transportation to and from the dialysis facility over symptoms related to underdialysis. Engaging patients and families in discussions often leads to identification of the issues and symptoms that lead to a request to reduce dialysis. Some patients will feel stressed when rushing to prepare for an early morning dialysis session or when they have to change their routine to eat their midday meal earlier and their evening meal later when attending afternoon dialysis. Dosing and timing of dialysis can then be adjusted accordingly to improve dialysis tolerance. A number of symptoms can also be mitigated by more frequent, but shorter, dialysis sessions, such as sleep disorders, respiratory distress, and postdialysis fatigue. This option may be acceptable for some patients who are not distressed by the dialysis procedure itself.

CONCLUSION

Care of the older patient with CKD has increasingly become a subspecialty in nephrology due to the challenge of balancing efficient, high-quality medical care with a personalized and humane approach to the older individual. Multiple areas of increased information and knowledge have been emerging in the field of geriatric nephrology, heralding a new era of personalized quality care. At the core is appreciation of the need for a better and deeper understanding of the individual to whom we are providing care before offering treatment suggestions or evaluations.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. An 82-year-old woman of Asian heritage is referred to your chronic kidney disease clinic for evaluation of a high creatinine level. She has a past medical history of hypertension for 42 years, treated only with a thiazide diuretic (chlorthalidone, 12.5 mg) and, recently a small dose of ACE inhibitor (ACEI; ramipril, 5 mg); osteoarthritis affecting her knees and hips; and glaucoma. She has no systemic symptoms, and her medical history is negative for cardiac or cerebral vascular disease or falls. She had three full-term pregnancies in her early 20s.

   On examination, she attends clinic alone and is oriented and looking well. She walks into the clinic room using a single-point cane and does not require assistance with getting onto the examination couch—weight, 51 kg; height, 153 cm. BP is controlled at 132/76 mm Hg lying, 132/74 mm Hg sitting, and 124/64 mm Hg standing. She denies dizziness on standing. No other clinical abnormalities are noted.

   In addition to the thiazide she is taking vitamin D, 400 IU/day, and eyedrops for her glaucoma. She is not taking any over-the-counter (OTC) medications. Creatinine level at presentation is 2.2 mg/dL, urinalysis is negative for proteinuria; complete blood count (CBC), electrolytes, albumin, and extended electrolytes are within normal limits, and the plasma protein electrophoresis is negative. A renal ultrasound shows mildly reduced renal size (8.7 and 9.1 cm for R and L kidney, respectively) and cortical thinning. Which of the following statements is true?

   a. Were she to start dialysis, her risk of mortality would be greater than 50% within the first 3 months of treatment.
   b. She is at increased risk of falls due to the change in blood pressure on standing.
   c. Her risk of mortality outweighs the risk of renal deterioration to the point of dialysis requirement.
   d. Use of the BIS equation to estimate eGFR is appropriate.

   Answer: c

   Rationale: Her risk of mortality outweighs the risk of renal deterioration to the point of dialysis requirement. Statement a is false; it is closer to 15%– 25%. Statement b is false; postural hypotension is defined as a systolic drip of 20 mm Hg or more. Statement d is false; use of the BIS equation has not yet been validated in nonwhite individuals.

2. You are asked to assume care for a 78-year-old man who has been maintained on dialysis for the past 27 months. He is dialyzed via a left forearm graft, with good adequacy. His dialysis history appears unremarkable, apart from two graft thromboses 14 and 18 months previously. His past medical history is positive for vascular renal disease, mild peripheral vascular disease (claudication on walking 250 m), a myocardial infarction 3½ years previously, and a three-vessel coronary artery bypass surgery 2½ years ago. He had one episode consistent with a transient ischemic attack approximately 7 months previously for which the workup was negative. Apart from chronic sleep issues that have not changed for several years, and long-standing pruritus, he denies any new symptoms. He is able to confirm he is taking each one of his medications as listed, but reports not always taking the sevelamer.

   You meet him in the dialysis unit and note that he appears to be comfortable at rest, with minimal respiratory discomfort. There is also a walker placed to the side of the dialysis machine that the patient confirms is his walker. He has brought some food with him and is eating a sandwich when you meet. He seems keen to continue eating because he reports he is extremely hungry. Vitals are within normal limits, although you note his postdialysis blood pressure is often 35 mm Hg lower than at the time of starting dialysis. Examination of cardiovascular and respiratory systems (performed ≈25 minutes into the dialysis treatment) reveals mild edema and a few scattered crepitations in both the right and left lung fields. A neurologic examination is not formally completed at this meeting but you note some resting tremor.

   His medications include intravenous erythropoietin, 3000 IU × 3/week; activated vitamin D, 0.25 μg/day; sevelamer, 1600 mg tid with food; pantoprazole, once daily; renal multivitamins; aspirin, enteric-coated; atorvastatin, 40 mg daily; metoclopramide, 10 mg tid as needed; ezetimibe, 10 mg daily; bisoprolol, 5 mg daily; amlodipine, 5 mg daily; midodrine, 10 mg daily; hydroxyzine, 25 mg tid; vitamin C, 100 IU daily; vitamin D, 400 IU daily; lorazepam, 2 mg at night; melatonin, 9 mg at night; mirtazapine, 15 mg at night; salbutamol, 2 puffs bid; senna, 2 tablets bid; lactulose, 15 mL as needed; Nitrospray, as needed. Which one of the following will you prioritize based on your initial geriatric assessment?

   a. You prioritize referral to an occupational therapist for evaluation of cognition because he is manifesting early signs of dementia and is at high risk of further deterioration.
   b. There is polypharmacy, and rationalization of the medications in collaboration with a pharmacist may reduce fall risk and cognitive decline and improve well-being.
   c. The significant drop in blood pressure postdialysis is of concern, and you arrange for assessment by a physiotherapist.
   d. The presence of the tremor and the use of a walker suggest that this patient is at increased risk of falls, and you recommend he urgently undergoes evaluation at the falls clinic.

   Answer: b

   Rationale: There are a significant number of medications that could be discontinued if no other indication is found. Examples include the pantoprazole, mirtazapine, and antihistamine. The antihistamine is not controlling the itch and therefore it is appropriate to have a trial off the drug. Statement a is false; the definition of dementia includes a functional component. The history given suggests that the patient's function remains intact; although further comment on cognitive function cannot be made at this time, a priority assessment by an OT is not likely needed. Statement c is false; although a postdialysis drop in BP is seen, the patient is asymptomatic and is not meeting the definition for postural hypotension. A careful observation of postdialysis BP is warranted and perhaps reevaluation of the timing of the BP agents. Statement d is false. Although both the presence of a tremor and the need for a walker suggest some parkinsonian features, as an initial step this warrants further assessment when the patient is off dialysis. The vascular history places this patient at high risk of vascular parkinsonism. Assessment of gait limitations and use of the walking aid, by a physiotherapist, may be useful in the near future.

3. Mr. TP, a 79-year-old man, is admitted to hospital after a fall precipitated by an episode of rapid atrial fibrillation. He has sustained a sprain injury of the left elbow and bruising around the left side of his face. Mr. TP is undergoing hemodialysis, having started dialysis 19 months ago after being admitted with fluid overload in association with ischemic cardiomyopathy and chronic kidney disease. There is a remote history of heavy alcohol intake. He and his wife moved into his daughter's home some months ago. This was precipitated by an episode when his wife, who has been diagnosed with dementia, was found wandering outside the home looking for him during one of his dialysis sessions. He supervises his wife's care, but he himself has been requiring in-home support for bathing for some months. Although mobile with a walking stick, he is accompanied to dialysis by his grandson who attends college nearby. His daughter provides support with most independent activities of daily living (iADLs) and with medication monitoring for both parents. He has been admitted to hospital on three occasions in the past 12 months with pneumonia (10 days), a viral gastrointestinal infection (8 days) and, most recently, with an unwitnessed fall, resulting in facial bruising and a 4-cm scalp laceration (3 days). Since then, the daughter drives her father home after dialysis.

   On admission, he is noted to have persistent atrial fibrillation. His CHADS2 score is 2 and HASBLED score is 4. What is your approach to the management of his atrial fibrillation?

   a. There are clear contraindications to anticoagulation, and therefore shared decision making is not appropriate.
   b. The benefit-risk profile is controversial. A discussion with the patient and his daughter, as his caregiver, is required. The clinician is responsible for providing all the available treatment options but must be careful to not guide the choice.
   c. Guidelines state that this man should be anticoagulated. As the data suggest, he would have more benefit than harm; the shared decision making discussion should emphasize the overall need for anticoagulation and focus only on the benefits.
   d. The benefit-risk profile is controversial but favor advising against anticoagulation because of the increasing number of geriatric issues over the past few months. A discussion with the patient (and his daughter if he is agreeable) is required.

   Answer: d

   Rationale: The factors against anticoagulation include doubt in the literature about the overall benefits of anticoagulation balanced against an apparent increased risk in this man. The information given suggests that this man is either frail or prefrail (suggested by recurrent admissions for minor illnesses, declining social functioning, and the recent fall). Statement a is false. Although there are clearly higher risks of bleeding than usual, and you may favor one approach over the other, shared decision making is always appropriate. Statement b is false. The clinician is responsible for providing all the appropriate treatment options in the context of the individual's clinical condition. Although they are not entitled to guide the choice, the clinician should help the patient and family establish realistic patient goals and then align these goals with the most likely treatment outcomes. Statement c is false. Although, overall, the population risk favors the use of anticoagulation for atrial fibrillation, it is important to assess the risk for each individual patient. In all scenarios, both benefits and risks should be discussed during shared decision making discussions.

4. Ms. LR, a 76-year-old woman, is brought to the hospital by her niece for symptoms of fatigue, nausea, poor appetite, and confusion for the past 2-3 weeks. Her laboratory tests show a serum creatinine level of 5.6 mg/dL, bicarbonate, 16 mEq/L, and potassium, 5.4, mEq/L. Her serum creatinine level 2 years ago was 3.1 mg/dL but she has not been seen by a physician for some years. She has a long history of hypertension managed with two antihypertensive medications, is not diabetic, and has had no other medical history of note. You are asked by the attending physician to initiate dialysis for Ms. LR's uremic symptoms and advanced kidney disease.

   You complete a review of history, clinical examination, and assessment. On examination, vitals are stable; however, she is unwell, with signs suggestive of mild volume depletion. She is clearly not able to participate actively in discussions. You then meet with the patient's niece who has power of attorney for personal care. The niece appears educated and engaged and clearly has the patient's best interest at heart. She has had previous nursing training but is now working in the pharmaceutical industry. The niece tells you that her aunt was an active lady who used to manage her own affairs and take care of herself until approximately 18 months ago. At that time, Ms. LR's husband died, and the niece noted a gradual decline in her aunt's well-being and functional independence. She reports that her aunt participates less in conversations and is more forgetful. The niece thinks that her aunt has been depressed since her uncle's death, and has tried on several occasions to discuss this with her aunt. She refused to see her family physician and has actively avoided contact with any of the health care team. Until recently, the niece says she telephoned her aunt weekly but visited only once per month. Six weeks ago, however the niece got an emergency call from the neighbors after they witnessed Ms. LR trip and fall to the ground just outside her home. At that time, the niece noted the house to be cluttered and messy and, since then, she has been visiting more frequently. At present, Ms. LR has difficulty with bathing and dressing, so her niece comes twice weekly to give her a full bath and bring some frozen meals, which she barely eats. She has lost some weight over the last year. Ms. LR's mobility is impaired due to chronic knee and back pain and she is afraid of falling, so prefers to stay seated on the couch rather than in bed. She uses a rollator walker inside the house and rarely goes outside. Her niece reports that all instrumental activities of daily living are now performed by the niece or the patient's sister.

The niece is familiar with dialysis and is not sure it would be "a good thing" for her aunt because she prefers to stay home as much as possible and avoid hospital visits. She says that with her aunt's back and knee pain, her aunt would probably not tolerate being seated on a dialysis chair for prolonged periods. The niece tells you that she herself is very busy and is worried that the transportation to the dialysis unit three times a week will be very challenging. She wants to know what her aunt's remaining life expectancy will be if she were to start dialysis and how her aunt would benefit from this treatment in terms of longevity. What information do you give to the niece?

a. You explain that dialysis is indicated because her aunt is acidemic, hyperkalemic, and symptomatic from uremia. You remind the niece of her role as the substitute decision maker and that her aunt will die without dialysis. You advise her that you expect her aunt's appetite and confusion will improve with dialysis and expect that she will also feel better. You estimate the median life expectancy to be close to 5 years.
b. You explain that dialysis is indicated because her aunt is acidemic, hyperkalemic, and symptomatic from uremia. You estimate the median survival to be around 3.5 years. You remind the niece of her role as the substitute decision maker and that her aunt will die without dialysis.
c. You explain that dialysis is indicated because her aunt is acidemic, hyperkalemic, and symptomatic from uremia. You estimate the median survival to be around 3.5 years but also state that the prognosis may be affected by the presence of several geriatric syndromes. You mention that she is at risk of further functional and cognitive decline despite dialysis.
d. You explain that dialysis is unlikely to offer much under these circumstances. Her median life expectancy with dialysis would be close to 1 year if she were to have had a fistula in place. You agree that dialysis care may not be appropriate and suggest referral to a palliative care unit nearby.

Answer: c

Rationale: The history is suggestive of several geriatric syndromes, including a recent fall, cognitive impairment, depression, a frailty phenotype, and malnutrition. Although symptoms such as nausea or confusion may improve with dialysis, others such as balance, depression, energy, and well-being may worsen. The decision around dialysis will depend on the patient's goals and may be influenced by the observation that she actively avoided contact with health care workers.